03 Current Electricity

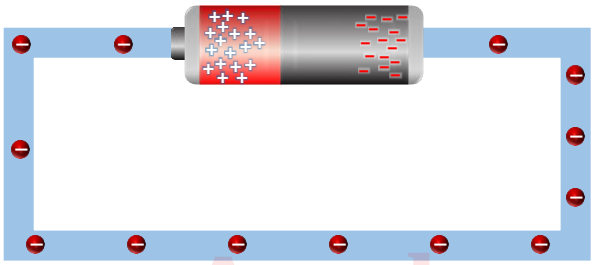
Current Microscopic Analysis

Electric Current

Time rate of flow of charge through a cross sectional area is called Current. If Δq charge flows in time interval Δt then average current is given by

$$I_{av} = \frac{\Delta q}{\Delta t}$$

Instantaneous current; $I = \lim_{\Delta t \to 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt}$



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⟵ Direction of electron flow

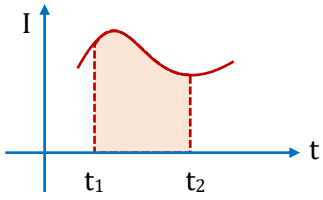
(Conventionally, direction of current is shown from positive to negative)

Direction of current is along the direction of flow of positive charge or opposite to the direction of flow of negative charge. But the current is a scalar quantity.

$\longrightarrow I$ $\qquad$ $I \longleftarrow$

$q\oplus \longrightarrow$ velocity $\qquad$ $q\ominus \longrightarrow$ velocity

- SI unit of current is ampere and 1 Ampere = 1 coulomb/sec
- It is a scalar quantity because it does not obey the law of vectors.
- Area under *I*-*t* curve provides total charge flow.



$$\text{Area} = \int_{t_1}^{t_2} I dt = \Delta q$$

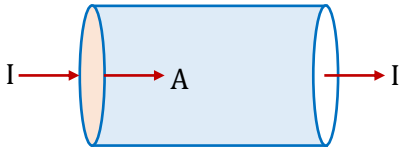
Current, velocity and current density

Current density at any point inside a conductor is defined as a vector having magnitude equal to current per unit area surrounding that point. Remember area is normal to the direction of charge flow (or current passes) through that point.

- Current density $\vec{J}$ is a vector quantity. It's direction is same as that of $\vec{E}$. It's S.I. unit is ampere/m^2 and dimensions $[L^{-2}A]$.

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- Current density at point P is given by $\vec{J} = \frac{dI}{dA}\vec{n}$



- If the cross–sectional area is not normal to the current, but makes an angle θ with the direction of current then

$$J = \frac{dI}{dA\cos\theta} \Rightarrow dI = JdA\cos\theta = \vec{J}.d\vec{A} \Rightarrow I = \int \vec{J}.\overrightarrow{dA}$$

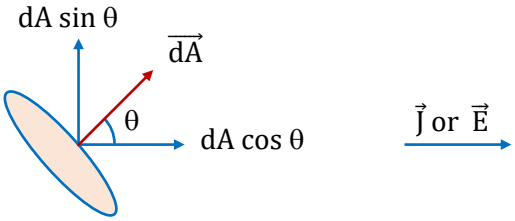


Illustration 1:

Current through a wire decreases uniformly from 4 A to zero in 10 s. Calculate charge flown through the wire during this interval of time.

Solution:

Charge flown = average current × time

$$= \left[\frac{4+0}{2}\right] \times 10 = 20C$$

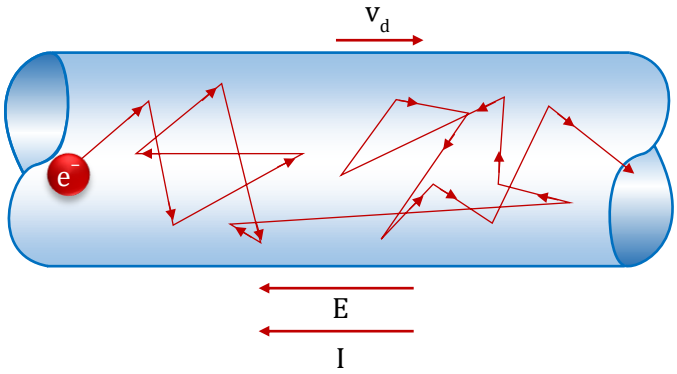
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Movement of Electrons Inside Conductor

All the free electrons are in random motion due to the thermal energy and relationship is given by

$$\frac{3}{2}KT = \frac{1}{2}mv^2$$

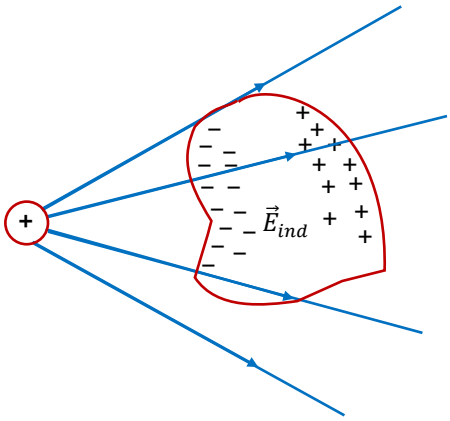
At room temperature its speed is around 10^6 m/sec or 10^3 km/sec



But the average velocity is zero so current in any direction is zero.

When a conductor is placed in an electric field, then for a small duration, electron do have an average velocity but its average velocity becomes zero within short interval of time.

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Mean Free Path (λ)

The distance travelled by a conduction electron during relaxation time is known as mean free path λ.

Mean free path of conduction electron = Thermal velocity × Relaxation time

$$\lambda = \frac{\lambda_1 + \lambda_2 \ldots . + \lambda_N}{N}$$

Order of λ = 10Å.

Relaxation Time (τ)

It is defined as average time elapsed between two successive collisions.

It is of the order of 10^{-14} s. It is a temperature dependent characteristic of the material of the conductor. It decreases with increase in temperature.

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$$\tau = \frac{\tau_1 + \tau_2 + \tau_3 + \ldots . \tau_N}{N}$$

Thermal Speed

Conductor contains a large number of free electrons, which are in continuous random motion.

Due to random motion, the free electrons collide with positive metal ions with high frequency and undergo change in direction at each collision. So, the thermal velocities are randomly distributed in all possible directions.

$\vec{u}_1, \vec{u}_2, \ldots \vec{u}_N$ are the individual thermal velocities of the free electrons at any given time.

The total number of free electrons in the conductor = N

$$\text{Average velocity } \vec{u}_{avg} = \left[\frac{\vec{u}_1 + \vec{u}_2 + \ldots \vec{u}_N}{N}\right] = 0$$

The average velocity is zero but average speed is non-zero.

Illustration 2:

The current through a wire depends on time as $i = i_0 + \alpha \sin \pi t$, where $i_0 = 10\ A$ and $\alpha = \frac{\pi}{2} A$. Find the charge crossed through a section of the wire in 3 seconds, and average current for that interval.

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Solution:

$$I = \frac{dq}{dt}$$

$$dq = Idt$$

$$q = \int_0^3 (i_0 + \alpha \sin \pi t)dt$$

$$= \left[i_0 t + \frac{\alpha}{\pi}(-\cos \pi t)\right]_0^3$$

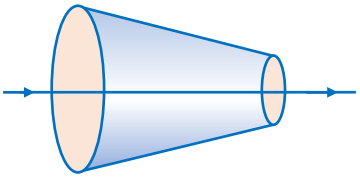
$$q = 3i_0 + \frac{2\alpha}{\pi} = 31C$$

Average current is :

$$I_{avg} = \frac{q}{\Delta t} = \frac{31C}{3s} \quad \Rightarrow \quad I_{avg} = \frac{31}{3}A$$

Illustration 3:

For non-uniform cross-sectional area compare current density at 1 and 2 cross section



Solution:

$$J_1 = \frac{I}{A_1}; J_2 = \frac{I}{A_2}; \ A_2 < A_1 \Rightarrow J_1 < J_2$$

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Drift Velocity ($\vec{V}_d$)

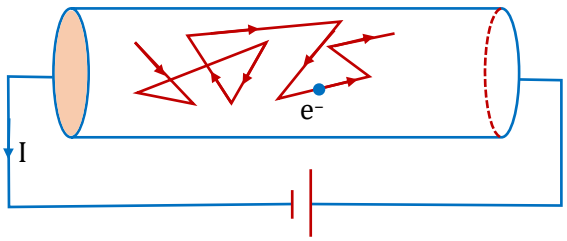
Drift velocity is defined as the velocity with which the free electrons get drifted towards the positive terminal under the effect of the applied electric field.

When the ends of a conductor are connected to a source of emf, an electric field E is established in the conductor, such that $E = \frac{V}{\ell}$

Where V = the potential difference across the conductor and ℓ= the length of the conductor.

The electric field $\vec{E}$exerts an electrostatic force$-e\vec{E}$ on each electron in the conductor.

The acceleration of each electron $\vec{a} = \frac{-e\vec{E}}{m}$



Under the action of electric field:
Random motion of an electron
with superimposed drift

m = mass of electron

e = charge of electron

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In addition to its thermal velocity, due to this acceleration, the electron acquires, a velocity component in a direction opposite to the direction of the electric field.

The gain in velocity due to the applied field is very small and is lost in the next collision.

At any given time, an electron has a velocity, $\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1$

Where, $\vec{u}_1$ = the thermal velocity

$\vec{a}\tau_1$ = the velocity acquired by the electron under the influence of the applied electric field.

τ_1 = the time that has elapsed since the last collision.

Similarly, the velocities of the other electrons are

$$\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2, \vec{v}_3 = \vec{u}_3 + \vec{a}\tau_3, \ldots, \vec{v}_N = \vec{u}_N + \vec{a}\tau_N$$

The average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v_d}$ of the free electrons.

$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \ldots \vec{v}_N}{N} = \frac{(\vec{u}_1 + \vec{a}\tau_1) + (\vec{u}_2 + \vec{a}\tau_2) + \ldots + (\vec{u}_N + \vec{a}\tau_N)}{N}$$

or $\vec{v}_d = \frac{(\vec{u}_1 + \vec{u}_2 + \ldots \vec{u}_N)}{N} + \vec{a}\frac{(\tau_1 + \tau_2 + \ldots + \tau_N)}{N}$ order of drift velocity is 10^{-4} m/s

$\because$ $\frac{\vec{u}_1 + \vec{u}_2 + \ldots + \vec{u}_N}{N} = 0$

$\therefore$ $\vec{v}_d = \vec{a}\frac{\tau_1 + \tau_2 + \ldots + \tau_N}{N} r$

$\Rightarrow$ $\vec{v}_d = \frac{e\vec{E}}{m}\tau$

Mobility

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As we have seen, conductivity arises as a result of mobile charge carriers. In metals, these mobile charge carriers are electrons; in an ionised gas, they are electrons and positively charged ions; in an electrolyte, these can be both positive and negative ions.

An important quantity is the mobility μ defined as the magnitude of the drift velocity per unit electric field:

$$\mu = \frac{|v_d|}{E}$$

The SI unit of mobility is $m^2/V\text{-}s$ and is 10^4 times the mobility in practical units ($cm^2/V\text{-}s$). Mobility is positive.

$$v_d = \frac{e\tau}{m}E \quad \Rightarrow \mu = \frac{V_d}{E} = \frac{e\tau}{m}$$

Where τ is the relaxation time for electrons.

Note:

On increasing temperature of conductor, frequency of collision increases and hence average relaxation time decreases.

Temperature ↑, τ ↓

$V_d = \frac{eE\tau}{m}, V_d \downarrow$

$\sigma = \frac{ne^2\tau}{m}, \sigma \downarrow, \rho \uparrow$ (Where ρ is resistivity of the material.)

$\mu = \frac{e\tau}{m}, \mu \downarrow$

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Illustration 4:

A current of 1.34 A exists in a copper wire of cross-section area 1.0 mm^2. Assuming each copper atom contributes one free electron. Calculate the drift speed of the free electrons in the wire. The density of copper is 8990kg/m^3 and atomic mass = 63.50.

Solution:

Mass of $1m^3$ volume of the copper is = 8990 kg

$= 8990 \times 10^3$ g

Number of moles in $1m^3 = \frac{8990\times10^3}{63.5} = 1.4 \times 10^5$

Since each mole contains 6×10^{23} atoms therefore number of atoms in 1m^3

$n = (1.4 \times 10^5) \times (6 \times 10^{23}) = 8.4 \times 10^{28}$ = electron density

$i = neAv_d$

$v_d = \frac{i}{neA} = \frac{1.34}{8.4\times10^{28}\times1.6\times10^{-19}\times10^{-6}}$ $(\because 1mm^2 = 10^{-6}m^2) = 10^{-4}m/s$

Illustration 5:

Two wires each of radius of cross-section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio:

(A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

Solution:

$I = neAv_d$

$\Rightarrow \quad v_d \propto \frac{1}{n}$

$\Rightarrow \quad \frac{v_{d_1}}{v_{d_2}} = \frac{n_2}{n_1} = \frac{4n_1}{n_1} = 4 : 1$

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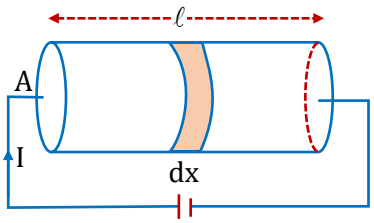
Ohm's Law

Let the number of free electrons per unit volume in a conductor = n

Total number of electrons in dx distance = n (Adx)

Total charge, $dQ = n\,(Adx)e$

Cross sectional area = A



Current $I = \frac{dQ}{dt} = nAe\frac{dx}{dt}$ $\Rightarrow$ $I = neAv_d$

Current density $J = \frac{I}{A} = nev_d$ $\Rightarrow$ $J = ne\left(\frac{eE}{m}\right)\tau$

$v_d = \left(\frac{eE}{m}\right)\tau$

$J = \left(\frac{ne^2\tau}{m}\right)E$ $\Rightarrow$ $J = \sigma E$, σ is called conductivity $(\sigma) = \frac{ne^2\tau}{m}$

In vector form $\vec{J} = \sigma\vec{E}$

σ depends only on the material of the conductor and its temperature.

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As temperature $(T) \uparrow$, $\tau \downarrow$

Now $J = \frac{I}{A}$, $\sigma = \frac{ne^2\tau}{m}$ and $E = \frac{V}{\ell}$

$$I = \frac{nAe^2\tau}{m\ell}V$$

$$V = \frac{m\ell}{nAe^2\tau}I$$

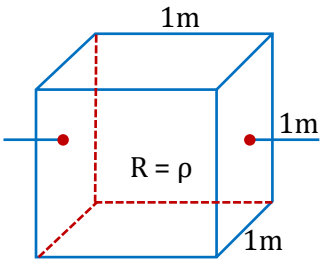
$$V = IR$$

So current in conductors is proportional to potential difference applied across its ends. This is Ohm's Law. R has S.I unit ohm (Ω).

Resistivity

It is the property of substance, defined as

$\rho = \frac{RA}{\ell}$, if ℓ = 1m, A = 1m² then $\rho = R$



It's unit is Ω-m

The specific resistance of a material is equal to the resistance of the wire of that material with unit cross – section area and unit length.

Resistivity depends on:

(i) Nature of material

(ii) Temperature of material

ρ does not depend on the size and shape of the material because it is the characteristic property of the conductor material.

$\rho_{alloy} > \rho_{semiconductor} > \rho_{conductor}$

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Conductivity

It is the measure of the ease at which an electric charge can pass through a material.

It's SI unit is Siemens per meter (S/m)

It is the inverse of resistivity $\sigma = \frac{1}{\rho}$

Limitations of Ohm's Law

Although Ohm's law has been found valid over a large class of materials, there do exist materials and devices used in electric circuits where the proportionality of V and I does not hold. The deviations broadly are one or more of the following types:

(a) V ceases to be proportional to I (Fig. 1).

(b) The relation between V and I depends on the sign of V. In other words, if I is the current for a certain V, then reversing the direction of V keeping its magnitude fixed, does not produce a current of the same magnitude as I in the opposite direction (Fig. 2). This happens, for example, in a diode

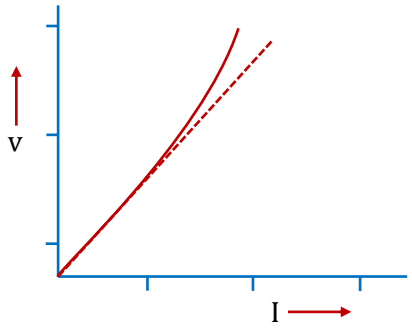


Fig-1 : The dashed line represents the linear ohm's law. The solid line is the voltage V versus current I for a good conductor.

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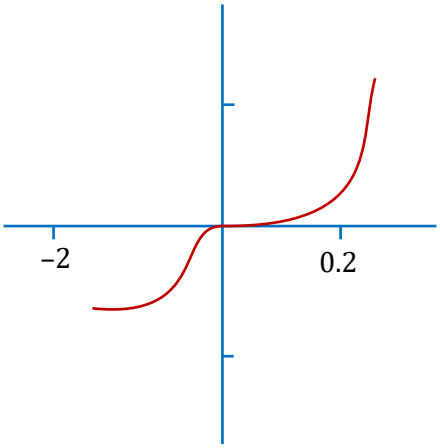


Fig -2 : Characteristics curve of a diode. Note the different scales for negative and positive values of the voltage and current.

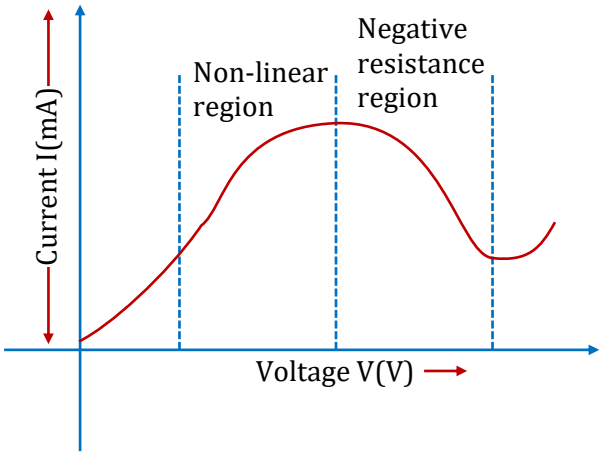


Fig.3 Variation of current versus voltage for GaAs.

(c) The relation between V and I is not unique, i.e., there is more than one value of V for the same current I (Fig. 3). A material exhibiting such behaviour is GaAs.

Materials and devices not obeying Ohm's law in the form $V = IR$, are actually widely used in electronic circuits.

Illustration 6:

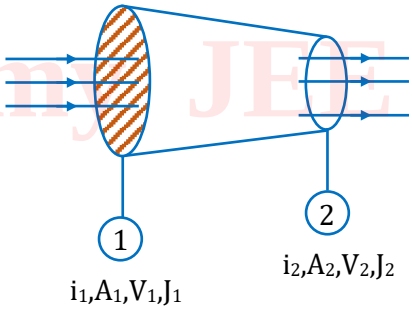
Current is flowing from a conductor of non-uniform cross section area if $A_1 > A_2$. Then find relation between

(a) i_1 and i_2

(b) J_1 and J_2

(c) V_1 and V_2 (drift velocity)

where i is current, J is current density and V is drift velocity.



Solution:

(a) i = Charge flowing through a cross-section per unit time.

$\therefore \quad i_1 = i_2$

(b) $J = \frac{i}{A}$

as $A_1 > A_2$ then $J_1 < J_2$

(c) $J = nev_d$

$v_d = \frac{J}{ne}$

as $J_1 < J_2$ then, $V_1 < V_2$

Resistance and it's Temperature Dependence

Electrical Resistance

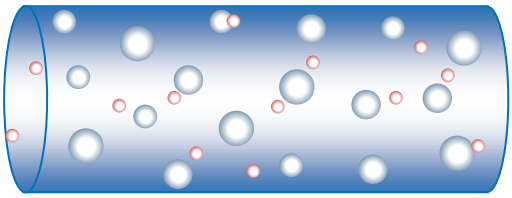
The property of a substance by virtue of which it opposes the flow of electric current through it is termed as electrical resistance. Electrical resistance depends on the size, geometry, temperature and internal structure of the conductor.

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We have, $I = \frac{nAe^2\tau}{m\ell}V$

$I = \frac{V}{R}$

$R = \frac{m\ell}{nAe^2\tau}$



Hence, $R = \frac{m}{ne^2\tau}\cdot\frac{\ell}{A}$

So, here $\boxed{R = \frac{\rho\ell}{A}}$

$\Rightarrow \quad V = I \times \frac{\rho\ell}{A}$

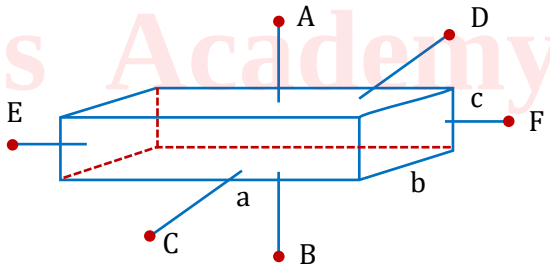
$\Rightarrow \quad \frac{V}{\ell} = \frac{I}{A}\rho \quad \Rightarrow \quad E = J\rho \quad \Rightarrow \quad J = \frac{I}{A}$ = current density

ρ is called resistively (it is also called specific resistance), and $\rho = \frac{m}{ne^2\tau} = \frac{1}{\sigma}$, σ is called conductivity. S.I. unit of resistivity is ohm-m (Ω-m). S.I. unit of conductivity is Ω^{-1}-m^{-1} also called siemens.

Illustration 7:

The dimensions of a conductor of specific resistance ρ are shown below. Find the resistance of the conductor across AB, CD and EF.

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Solution:

For a condition

$$R = \frac{\rho l}{A} = \frac{\text{Resistivity} \times \text{length}}{\text{Area of cross section}} \quad \Rightarrow R_{AB} = \frac{\rho c}{ab}, R_{CD} = \frac{\rho b}{ac}, R_{EF} = \frac{\rho a}{bc}$$

Dependence of Resistance on Various Factors

From previous discussion we have:

$$R = \rho\frac{\ell}{A} = \frac{m}{ne^2\tau} \times \frac{\ell}{A}$$

Therefore, resistance depends on

(1) length of the conductor ($R \propto \ell$)

(2) area of cross - section of the conductor $R \propto \frac{1}{A}$

(3) nature of material of the conductor $R = \frac{\rho\ell}{A}$

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Results:

(a) On stretching a wire (volume constant), if length of wire is taken into account then $\frac{R_1}{R_2} = \frac{\ell_1^2}{\ell_2^2}$

(b) If radius of cross section is taken into account then $\frac{R_1}{R_2} = \frac{r_2^4}{r_1^4}$, where R_1 and R_2 are initial and final resistances and ℓ_1, ℓ_2, are initial and final lengths and r_1 and r_2 initial and final radii respectively.

(c) Effect of percentage change in length of wire

$\frac{R_2}{R_1} = \frac{\ell^2[1+\frac{x}{100}]^2}{\ell^2}$, where ℓ - original length and x- % increment

If x is quite small (say < 5%) then % change in R is

$$\frac{R_2 - R_1}{R_1} \times 100 = \left(\frac{\left(1+\frac{x}{100}\right)^2 - 1}{1}\right) \times 100 \cong 2x\%$$

Illustration 8:

If a wire is stretched to double its length, find the new resistance if original resistance of the wire was R.

Solution:

As we know that $R = \frac{\rho\ell}{A}$ $\Rightarrow$ in case $R' = \frac{\rho\ell'}{A'}$

$\ell' = 2\ell$

$A'\ell' = A\ell$ (Volume of the wire remains constant)

$A' = \frac{A}{2}$ $\Rightarrow$ $R' = \frac{\rho\times 2\ell}{A/2} = 4\frac{\rho\ell}{A} = 4R$

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Illustration 9:

The wire is stretched to increase the length by 1%. Find the percentage change in the resistance.

Solution:

As we known that,

$\therefore$ $R = \frac{\rho\ell}{A}$

$\Rightarrow$ $\frac{\Delta R}{R} = \frac{\Delta\rho}{\rho} + \frac{\Delta\ell}{\ell} - \frac{\Delta A}{A}$ and $\frac{\Delta\ell}{\ell} = -\frac{\Delta A}{A}$

$\frac{\Delta R}{R} = 0 + 1 + 1 = 2$

Hence percentage increase in the Resistance = 2%.

Note:

Above method is applicable when % change is very small.

Temperature Dependence of Resistivity and Resistance:

The resistivity of a material is found to be dependent on the temperature over a limited range of temperatures (not too large). The resistivity of a metallic conductor is approximately given by:

$\rho_T = \rho_0(1 + \alpha(T - T_0))$

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Where ρ_T is the resistivity at a temperature T and ρ_0 is the resistivity at a reference temperature T_0. Here, α is called the temperature co-efficient of resistivity.

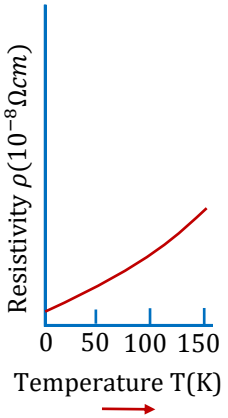


Figure - 1
Resistivity ρ_T of Copper as a function of temperature T.

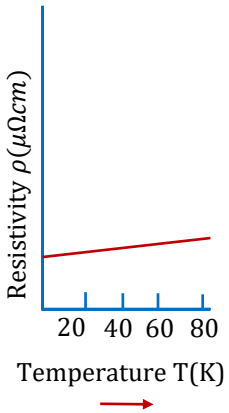


Figure - 2
Resistivity ρ_T of Nichrome as a function of absolute temperature T.

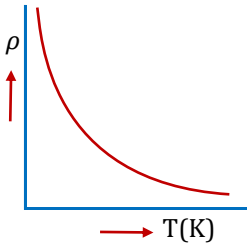


Figure - 3
Temperature dependence of resistivity for a typical semiconductor.

Unit of α is $°C^{-1}$.

Resistance corresponding to temperature difference (ΔT) is given as $R_t = R_0\,(1 + \alpha\Delta T)$

Where R_t = Resistance at $t°C$, R_0 = Resistance at 0°C

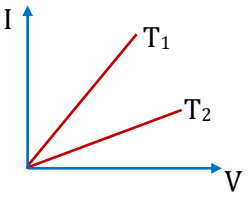
ΔT = Change in temperature, α = Temperature coefficient of resistance.

[For metals: α is positive but for semiconductors and insulators: α is negative]

Resistance of the conductor decreases linearly (figure-1) with decrease in temperature and becomes zero at a specific temperature. This temperature is called critical temperature. Below this temperature a conductor becomes a superconductor.

Illustration 10:

The current-voltage graphs for a given metallic wire at two different temperature T_1 and T_2 are shown in the figure. Which one is higher, T_1 or T_2.



Solution:

Slope of $I - V$ Curve is $\frac{1}{R}$

$$\left(\frac{1}{R}\right)_1 > \left(\frac{1}{R}\right)_2 \Rightarrow R_2 > R_1 \Rightarrow T_2 > T_1$$

Illustration 11:

The resistance of a thin silver wire is 1.0 Ω at 20°C. The wire is placed in liquid bath and its resistance rises to 1.2 Ω. What is the temperature of the bath? (Here $\alpha = 10^{-2}$ /°C)

Solution:

Here change in resistance is small so we can apply

$R = R_0(1 + \alpha\Delta\theta)$

$\Rightarrow$ $1.2 = 1 \times (1 + 10^{-2}\,\Delta\theta)$

$\Rightarrow$ $\Delta\theta = 20°C$

$\Rightarrow$ $\theta - 20 = 20$

$\Rightarrow$ $\theta = 40°\,C$ **Ans.**

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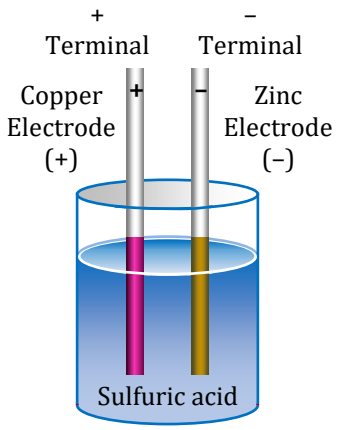
Cell, E.M.F and Internal Resistance

Battery (Cell)

A battery is a device which maintains a potential difference across its two terminals A and B. Dry cells, secondary cells, generator and thermocouple are the devices used for producing potential difference in an electric circuit. Arrangement of cell or battery is shown in figure.

Electrolyte provides continuity for current.

It is often prepared by putting two rods or plates of different metals in a chemical solution. Some internal mechanism exerts force ($\vec{F}_n$)on the ions (positive and negative) of the solution. This force drives positive ions towards positive terminal and negative ions towards negative terminal. As positive charge accumulates on anode and negative charge on cathode a potential difference and hence an electric field $\vec{E}$ is developed from anode to cathode. This electric field exerts an electrostatic force $\vec{F} = q\vec{E}$on the ions. This force is opposite to that of $\vec{F}_n$. In equilibrium (steady state) $F_n = F_e$ and no further accumulation of charge takes place.



Simple Electric Cell

When the terminals of the battery are connected by a conducting wire, an electric field is developed in the wire.

The free electrons in the wire move in the opposite direction and enter the battery at positive terminal. Some electrons are withdrawn from the negative terminal. Thus, potential difference and hence, F_e decreases in magnitude while F_n remains the same. Thus, there is a net force on the positive charge towards the positive terminal. With this the positive charge rush towards positive terminal and negative charge rush towards negative terminal. Thus, the potential difference between positive and negative terminal is maintained.

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Internal resistance (r):

The potential difference across a real source in a circuit is not equal to the emf of the cell. The reason is that charge moving through the electrolyte of the cell encounters resistance. We call this the internal resistance of the source.

Internal resistance is always connected in series with the source.

The internal resistance of a cell depends on the distance between electrodes ($r \propto d$), area of electrodes ($r \propto \frac{1}{s}$) and nature, concentration ($r \propto c$) and temperature of electrolyte ($r \propto \frac{1}{\text{Temp.}}$).

Illustration 12:

What is the meaning of 10 Amp. hr?

Solution:

It means if the 10 A current is withdrawn then the battery will work for 1 hour.

10 Amp $\longrightarrow$ 1 hr

1 Amp $\longrightarrow$ 10 hr

½ Amp $\longrightarrow$ 20 hr

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Electromotive force : (E.M.F)

Definition I : Electromotive force is the capability of the system to make the charge flow.

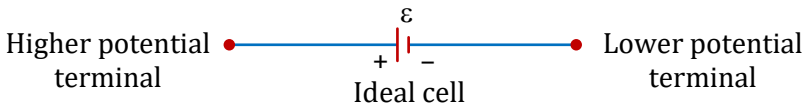
Definition II : It is the work done by the battery for the flow of 1 coloumb charge from lower potential terminal to higher potential terminal inside the battery.

Definition III : Electromotive force of a cell is equal to potential difference between its terminals when no current is passing through the circuit.

Representation for Battery:

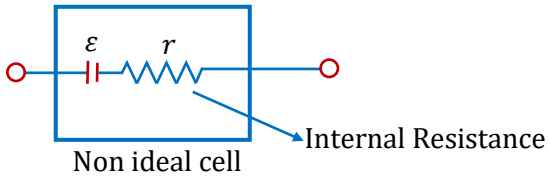
- **Ideal cell :**

  Cell in which there is no heating effect.



- **Non-ideal cell :**

  Cell in which there is heating effect inside due to opposition to the current flow internally



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Terminal Potential Difference (TPD or V)

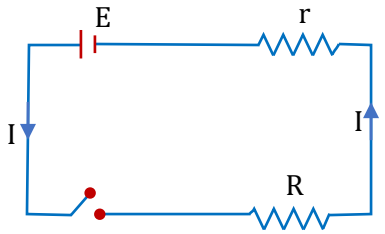
The potential difference between the two electrodes of a cell in a closed circuit i.e., when current is being drawn from the cell is called terminal potential difference.

(a) When cell is discharging:

When cell is discharging current inside the cell is from cathode to anode.

Current $I = \frac{E}{r+R}$ or $E = IR + Ir = V + Ir$ or $V = E - Ir$

When current is drawn from the cell potential difference is less than emf of cell. Greater is the current drawn from the cell smaller is the terminal voltage. When a large current is drawn from a cell its terminal voltage is reduced.

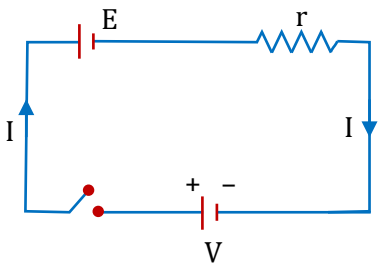


(b) When cell is charging:

When cell is charging current inside the cell is from anode to cathode.

Current $I = \frac{V-E}{r}$ or $V = E + Ir$

During charging terminal potential difference is greater than emf of cell.



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(c) When cell is in open circuit:

In open circuit, $R = \infty \quad \therefore \; I = \frac{e}{R+r} = 0 \quad$ So, $V = E$

In open circuit terminal potential difference is equal to emf and is the maximum potential difference which a cell can provide.

(d) When cell is short circuited:

In short circuit, $R = 0$ So, $I = \frac{E}{R+r} = \frac{E}{r}$ and $V = 0$

In short circuit, current from cell is maximum and terminal potential difference is zero.

Note:

- The potential at all points of a wire of zero resistance will be same.
- Earthing : If some point of circuit is earthed then its potential is assumed to be zero.

Important point

1. At the time of charging a cell. When current is supplied to the cell, the terminal voltage is greater than the e.m.f. E.
   $V = E + Ir$
2. Series combination is useful when internal resistance is less than external resistance of the cell.
3. Parallel combination is useful when internal resistance is greater than external resistance of the cell.
4. Power in R (given resistance) is maximum, if its value is equal to net resistance of remaining circuit.
5. Internal resistance of ideal cell = 0.
6. If external resistance is zero than current given by circuit is maximum.

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Value of External Resistance	Current from Cell	Terminal Potential Difference (TPD)
R	$I = \frac{E}{R+r}$	$V = E - Ir$
$R = 0$	$I = \frac{E}{r}$	$V = E - \frac{E}{r}r$
Short circuit	Maximum	$V = 0$
Open circuit $R = \infty$	$I = 0$	$V = E - 0$ $V = E$ $(TPD = EMF)$

Illustration 13:

By a cell a current of 0.9 A flows through 2 ohm resistor and 0.3 A through 7 ohm resistor. The internal resistance of the cell is

(A) 0.5Ω (B) 1.0 Ω (C) 1.2 Ω (D) 2.0 Ω

Solution:

$0.9 = \frac{\varepsilon}{(2+R)}$ ...(1)

$0.3 = \frac{\varepsilon}{(7+R)}$ ...(2)

On solving equation (1) and (2)

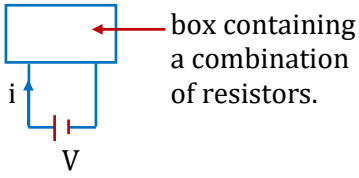
$R = 0.5\Omega$

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Series and Parallel Combination of Resistance

Combination of Resistances:

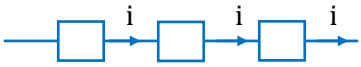
A number of resistances can be connected and all the complicated combinations can be reduced to two different types, namely series and parallel.



The equivalent resistance of a combination is defined as $R_{eq} = \frac{V}{i}$

Series Combination:

When the resistances are connected end to end then they are said to be in series. In series current through each element is same.



Resistances in series carry equal current but reverse may not be true.

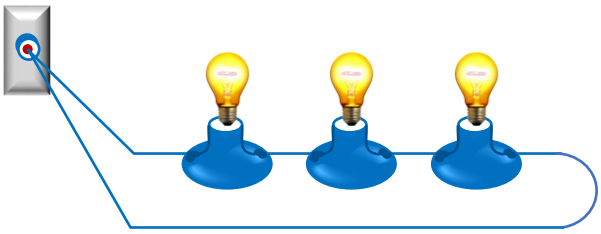
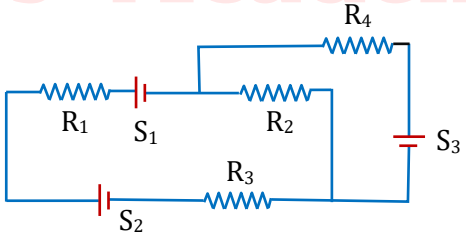


Illustration 14:

Which electrical elements are connected in series.

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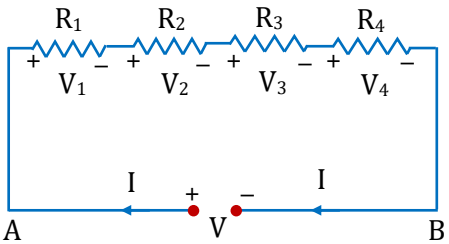


Solution:

Here S_1, S_2, R_1, R_3 connected in one series and R_4, S_3 connected in different series.

Equivalent Resistance of Series Combination

The potential difference across a resistor is proportional to the resistance.



$\because$ $V = IR_{eq}$ where I is same through any of the resistor.

and $V = V_1 + V_2 + V_3 + \ldots\ldots\ldots + V_n$ where

$$V_1 = \frac{R_1}{R_1+R_2+\ldots\ldots\ldots+R_n}V$$

$$V_2 = \frac{R_2}{R_1+R_2+\ldots\ldots\ldots+R_n}V \text{ and so on}$$

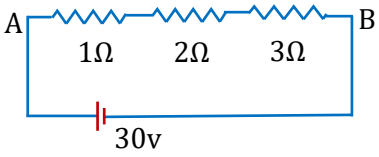
The effective resistance appearing across the battery (or between the terminals A and B) is

$R_{eq} = R_1 + R_2 + R_3 + \ldots\ldots\ldots + R_n$ (this means R_{eq} is greater than any resistor)

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Illustration 15:

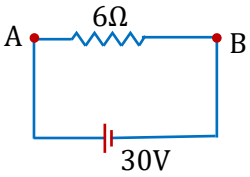
Find the current in the circuit.



Solution:

R_{eq} = 1 + 2 + 3 = 6Ω the given circuit is equivalent to

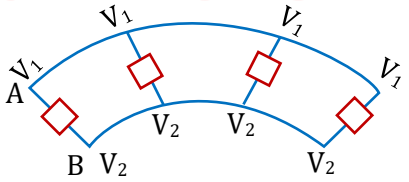
current $i = \frac{v}{R_{eq}} = \frac{30}{6} = 5A$ **Ans.**



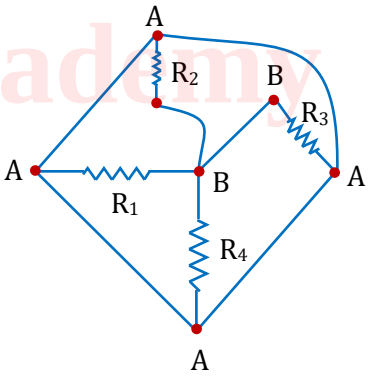
Parallel Combination:

A parallel circuit of resistors is one in which the same voltage is applied across all the components in a parallel grouping of resistors R_1, R_2, R_3, ..., R_n.

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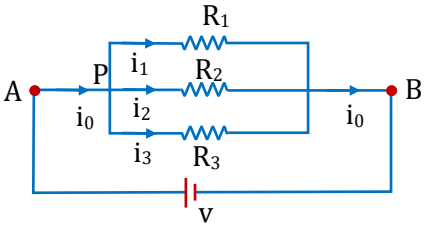
(a)



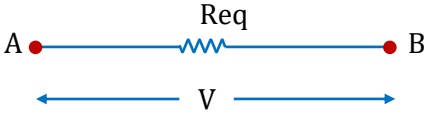
(b)

In the figure (a) and (b) all the resistors are connected between points *A* and *B* so they are in parallel.

Equivalent Resistance of Parallel Combination



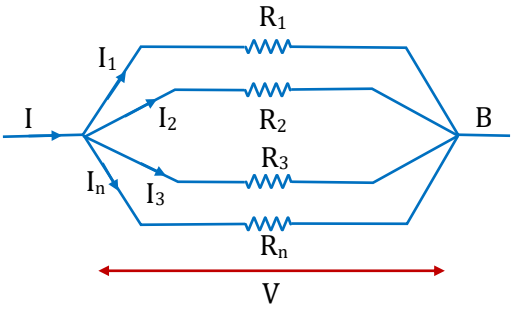
≡



At junction point *P*,

$i_0 = i_1 + i_2 + i_3$

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Therefore, $\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$

$\Rightarrow \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots\ldots + \frac{1}{R_n}$

Conclusions: (about parallel combination)

(a) Potential difference across each resistor is same.

(b) $I = I_1 + I_2 + I_3 + \ldots\ldots I_n$

(c) Effective resistance (R) then,

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

(R is less than each resistor).

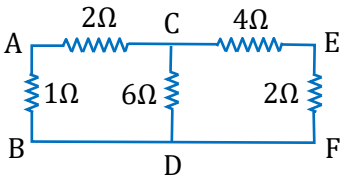
(d) Current in different resistors is inversely proportional to the resistance.

$I_1 : I_2 : \ldots : I_n = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3} : \ldots : \frac{1}{R_n}$

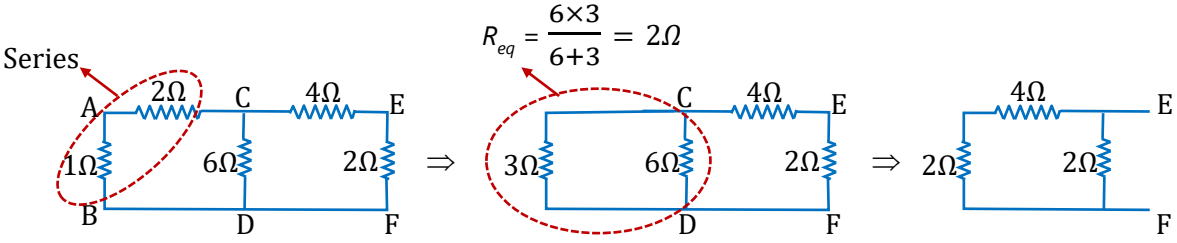
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Illustration 16:

Find the equivalent resistance of the circuit given in figure between E and F.



Solution:

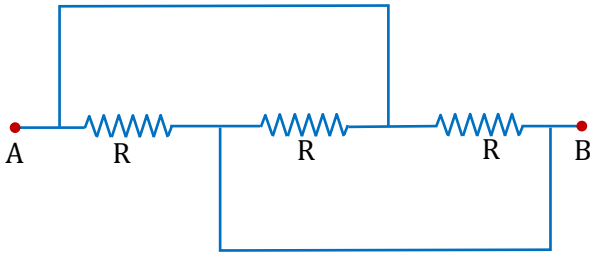


$\Rightarrow \quad R_{EF} = \frac{6\times2}{6+2} = \frac{3}{2}\Omega$

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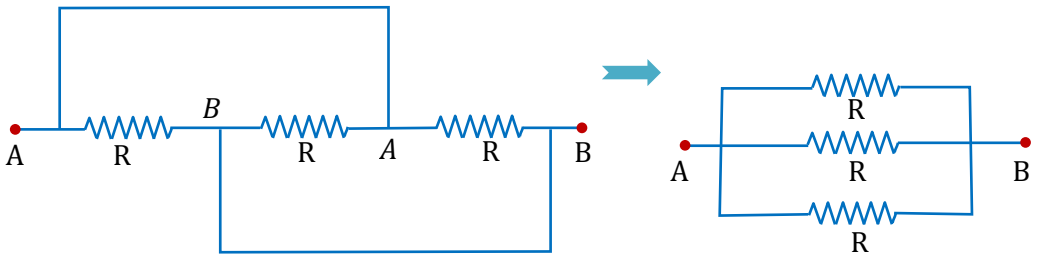
Illustration 17:

What is the equivalent resistance between A and B?



Solution:

All the three resistors can be considered in parallel as shown below:



$\therefore \quad R_{eq} = \frac{R}{3}$

Electric Power and Heating Effects of Current

Electrical Power:

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Energy liberated per second in a device is called its power. The electrical power P delivered or consumed by an electrical device is given by $P = Vi$, where V = Potential difference across the device and i = Current.

$$\text{Power } (P) = \frac{V.dq}{dt} = V\,i$$

If the current enters the higher potential point of the device then electric power is consumed by it (i.e. acts as load). If the current enters the lower potential point then the device supplies power (i.e. acts as source).



Heating Effect of Current

Cause of Heating:

The potential difference applied across the two ends of conductor sets up electric field. Under the effect of electric field, electrons accelerate and as they move, they collide against the ions and atoms in the conductor, the energy of e^- transferred to the atoms and ions appears as heat.

Joules's Law of Heating

Let voltage 'V' is applied across the conductor of resistance R and a charge dq flows through it in time dt then work done by the battery is

$$dw = dqV$$
$$= idt\,V$$
$$\int dw = \int i(iR)dt = \int i^2 R dt$$

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Since Heat energy released from the conductor is equal to work done by the battery unit,

H = Heat released = $W = \int i^2 R dt$

If i = constant, $H = i^2 Rt = \frac{V^2}{R} t = (Vi)t$

Heat produced in conductor does not depend upon the direction of current.

SI UNIT : joule ; 1 kWh = 3.6×10^6 joule = 1 unit ; 1 BTU (British Thermal Unit) = 1055 J

Power delivered to the load resistance

$$P = \frac{dw}{dt}$$

$$P = \frac{i^2 R dt}{dt} = i^2 R = \frac{V^2}{R} = Vi$$

Fusing of an Electrical Device

Every electrical device like bulb, heater, geyser, etc. comes with a design/rated voltage (V_0) and design/rated wattage or power (P_0).

If applied voltage (V) across a device becomes greater than V_0, then it will not work or we can say device will be fused.

Resistance of an electrical device can be found out using

$$P_0 = \frac{V_0^2}{R}$$

i.e $$R = \frac{V_0^2}{P_0}$$

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If V is the applied voltage across the device and if P is the power consumed by it, then

$$P = \frac{V^2}{R} i.e P = \left(\frac{V^2}{V_0^2}\right) P_0$$

Illustration 18:

If bulb rating is 100 watt and 220 V then determine:

(a) Resistance of filament

(b) Current through filament

(c) If bulb operate at 110-volt power supply then find power consumed by bulb.

Solution:

Bulb rating is 100 W and 220 V bulb means when 220 V potential difference is applied between the two ends then the power consumed is 100 W

Here, V = 220 Volt

P = 100 W

$\frac{V^2}{R}$ = 100 ; So, R = 484 Ω

Since Resistance depends only on material hence it is constant for bulb

$$I = \frac{V}{R} = \frac{220}{22 \times 22} = \frac{5}{11} \text{ Amp.}$$

Power consumed at 110 V

$\therefore$ Power consumed = $\frac{110 \times 110}{484}$ = 25 W

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Fuse wire

The fuse wire for an electric circuit is chosen keeping in view the value of safe current through the circuit.

The fuse wire should have high resistance per unit length and low melting point.

However, the melting point of the material of fuse wire should be above the temperature that will be reached on the passage of the current through the circuit.

A fuse wire is made of alloys of lead (Pb) and tin (Sn).

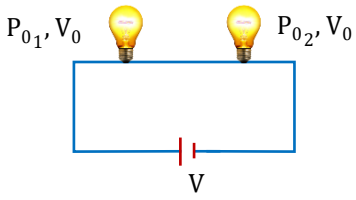
Length of fuse wire is immaterial.

The material of the filament of a heater should have high resistivity and high melting point.

Home Appliances (Bulb, Heater, Geyser)

Equivalent power in combination of Bulbs

(i) Series combination



Let, 2 bulbs with V_0 but $P_{0_1} > P_{0_2}$ are connected in series.

Here, $P_{0_1} > P_{0_2} \Rightarrow R_1 < R_2$

Now, since series $\Rightarrow P_1 = I^2R_1 \text{and} P_2 = I^2R_2$

$\therefore \quad P_1 < P_2$

In series combination of bulbs, since Brightness $\propto$ Power consumed by bulb $\propto R \propto \frac{1}{P_{\text{rated}}}$, bulb of lesser wattage will shine more.

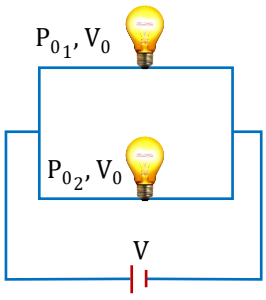
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P = total power of combination $= \frac{V^2}{R_1+R_2} = \frac{V^2}{\frac{V_0^2}{P_{0_1}}+\frac{V_0^2}{P_{0_2}}} = \left(\frac{V}{V_0}\right)^2 \left(\frac{P_{0_1}P_{0_2}}{P_{0_1}+P_{0_2}}\right)$

If $V = V_0$, then $\frac{1}{P} = \frac{1}{P_{0_1}} + \frac{1}{P_{0_2}}$

(ii) Parallel combination

Similarly, let two bulbs with V_0 but $P_{0_1} > P_{0_2}$ are connected in parallel.



Since, $P_{0_1} > P_{0_2}$

$\therefore \quad R_1 < R_2$

Now $P_1 = \frac{V^2}{R_1}; P_2 = \frac{V^2}{R_2}$

$\therefore \quad P_1 > P_2$

In parallel combination of bulbs since Brightness $\propto$ Power consumed by bulb $\propto I \propto \frac{1}{R} \propto P_{\text{rated}}$, so bulb of greater wattage will shine more.

$P_{\text{total}} = \frac{V^2}{R_1} + \frac{V^2}{R_2} = \left(\frac{V}{V_0}\right)^2 [P_1 + P_2]$

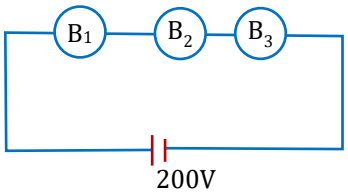
If $V = V_0$

$P_{\text{total}} = P_1 + P_2$

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Illustration 19:

In the figure shown B_1, B_2 and B_3 are three bulbs rated as (200V, 50 W), (200V, 100W) and (200 V, 25W) respectively. Find the current through each bulb and which bulb will give more light?



Solution:

$$R_1 = \frac{(200)^2}{50}; R_2 = \frac{(200)^2}{100}; R_3 = \frac{(200)^2}{25}$$

the current flowing through each bulb is

$$= \frac{200}{R_1+R_2+R_3} = \frac{200}{(200)^2\left[\frac{2+1+4}{100}\right]} = \frac{100}{200\times7} = \frac{1}{14}\text{A}$$

Since, $R_3 > R_1 > R_2$

$\therefore$ Power consumed by bulb = i^2R

$\therefore$ If the resistance is of higher value then it will give more light.

$\therefore$ Here Bulb B_3 will give more light.

Illustration 20:

A heater coil is rated 100 W, 200V. It is cut into two identical parts. Both parts are connected together in parallel, to the same source of 200V. Calculate the energy liberated per second in the new combination.

Solution:

$$P = \frac{V^2}{R}$$

$\therefore$ $R = \frac{V^2}{P} = \frac{(200)^2}{100} = 400\Omega$

Resistance of half piece = $\frac{400}{2} = 200\Omega$

Resistance of pieces connected in parallel = $\frac{200}{2} = 100\Omega$

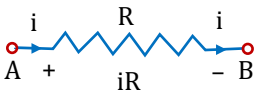
Energy liberated/second = $P = \frac{V^2}{R} = \frac{200\times200}{100} = 400\text{W}$

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Single Loop Circuit and Power Output, R for Maximum Power Output and Maximum Current

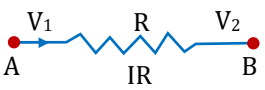
Electric current in resistance

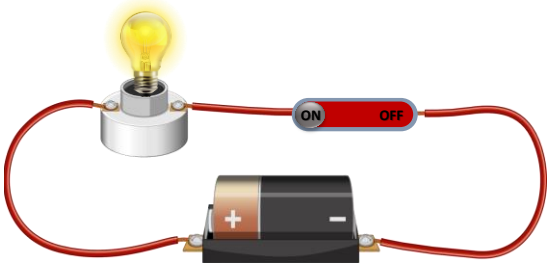
In a resistor current flows from high potential to low potential



High potential is represented by positive (+) sign and low potential is represented by negative (–) sign.

$V_A - V_B = iR$





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If $V_1 > V_2$

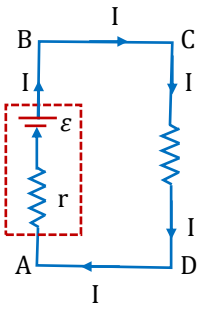
Then current will flow from A to B, and $i = \frac{V_1 - V_2}{R}$

If $V_1 < V_2$

Then current will go from B to A, and $i = \frac{V_2 - V_1}{R}$

Maximum Power Output

In the given electric circuit.



If we assume that potential at A is zero then potential at B is $\varepsilon - Ir$.

Now since the connecting wires are of zero resistance

$\therefore \quad V_D = V_A = 0 \quad \Rightarrow \quad V_C = V_B = \varepsilon - Ir$

Now current through CD is also I ($\because$ it's in series with the cell)

$\therefore \quad I = \frac{V_C - V_D}{R} = \frac{(\varepsilon - Ir) - 0}{R} \quad \Rightarrow \quad$ Current $I = \frac{\varepsilon}{r+R}$

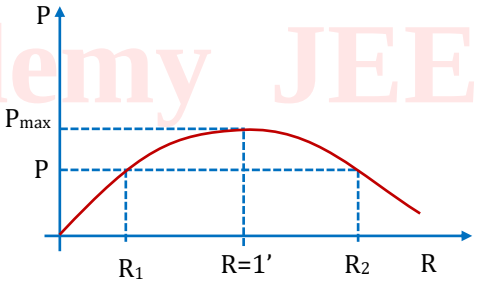
Power output in R $\quad \Rightarrow \quad P = I^2R = \frac{\varepsilon^2}{(r+R)^2} . R$

For maximum power supply $\quad \Rightarrow \quad \frac{dP}{dR} = \frac{\varepsilon^2}{(r+R)^2} - \frac{2\varepsilon^2 R}{(r+R)^3} = \frac{\varepsilon^2}{(R+r)^3}[R + r - 2R]$

For maximum power supply

$$\frac{dP}{dR} = 0 \Rightarrow r + R - 2R = 0 \Rightarrow r = R$$

Here for maximum power output external resistance should be equal to internal resistance.



Now, graph between 'P' and 'R'.

Maximum power output at R

$$P_{max} = \frac{\varepsilon^2}{4R} \quad \Rightarrow \quad I = \frac{\varepsilon}{r+R} = \frac{\varepsilon}{2R}$$

Now, from graph we can see that for a given power output there exists two values of external resistance.

$$P = \frac{\varepsilon^2 R}{(r+R)^2}$$

$$P(r^2 + 2rR + R^2) = \varepsilon^2 R$$

$$R^2 + \left(2r - \frac{\varepsilon^2}{P}\right)R + r^2 = 0$$

Above quadratic equation in R has two roots R_1 and R_2 for given values of ε, P and r such that

$\therefore \quad R_1R_2 = r^2$ (product of roots)

$r^2 = R_1R_2$

Illustration 21:

How will you connect (series and parallel) 24 cells each of internal resistance 1Ω to get maximum power output across a load of 10 Ω?

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Solution:

Suppose there are m rows and each row has n cells. The total number of cells is $mn = 24$.

Here $I = \frac{ne}{R+(nr/m)}$

For maximum power, internal resistance equals the load resistance

$$\frac{nr}{m} = R \text{ or } n = 10m \text{ or } 10m^2 = 24 \text{ or } m = \sqrt{2.4} = 1.55$$

As, m should be an integer. Therefore,

If $m = 1, n = 24, I = 24e/34$

then $P_1 = \left(\frac{24e}{34}\right) \times 10 = 4.98e^2$

If $m = 2, n = 12, I = 12e/16$

then $P_2 = \left(\frac{12e}{16}\right)^2 \times 10 = 5.625e^2$

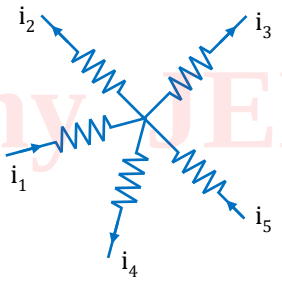
So, we have two rows ($m = 2$) each containing 12 cells ($n = 12$) in series.

Kirchhoff's Law

There are two laws given by Kirchhoff for determination of potential difference and current in different branches of any complicated network.

First law or junction rule

In an electric circuit, the algebraic sum of the current meeting at any junction in the circuit is zero.



or

Sum of the currents entering the Junction is equal to sum of the current leaving the Junction.

$$\sum i = 0$$

$$i_1 - i_2 - i_3 - i_4 + i_5 = 0 \Rightarrow i_1 + i_5 = i_2 + i_3 + i_4$$

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This is based on law of conservation of charge.

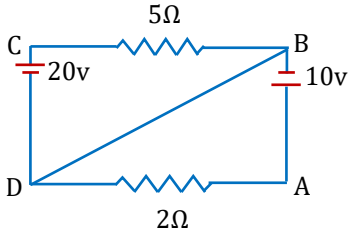
Relative Potential:

While solving an electric circuit it is convenient to choose a reference point and assigning its voltage as zero, then all other potentials are measured with respect to this point. This point is also called the common point.

Once potential has been allotted to each point, we write equations following Kirchhoff's Current Law (Outgoing Current through a point will be equal to the incoming current). These equations can be solved to give the potential at each point.

Illustration 22:

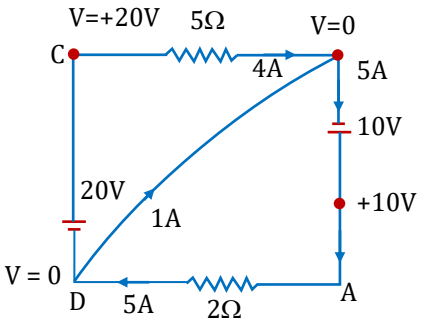
In the figure given beside find out the current in the wire BD



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Solution:

Let at point D potential $V = 0$ and write the potential of other points.



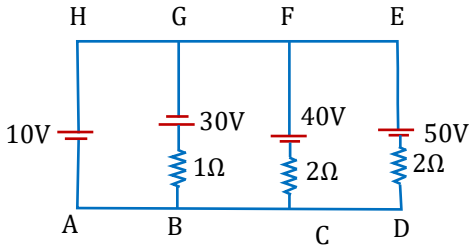
Then current in wire $AD = \frac{10}{2} = 5\ A$ from A to D.

Current in wire $CB = \frac{20}{5} = 4A$ from C to B.

$\therefore$ Current in wire $BD = 1\ A$ from D to B.

Illustration 23:

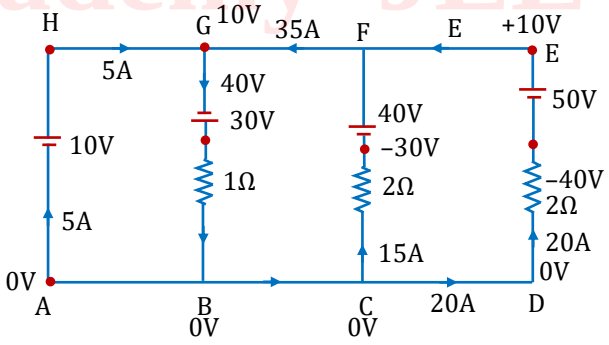
Find the current in each wire.



Solution:

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Let potential at point A is 0 volt then potential of other points is shown in figure.



Current in $BG = \frac{40-0}{1} = 40\ A$ from G to B

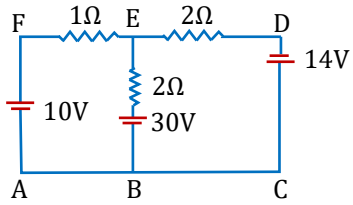
Current in $FC = \frac{0-(-30)}{2} = 15\ A$ from C to F

Current in $DE = \frac{0-(-40)}{2} = 20\ A$ from D to E

Current in wire $AH = 40 - 35 = 5\ A$ from A to H

Illustration 24:

Find the current in each wire.



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Solution:

Let potential at point $B = 0$. Then potential at other points are mentioned.

$\therefore$ Potential at E is not known numerically.

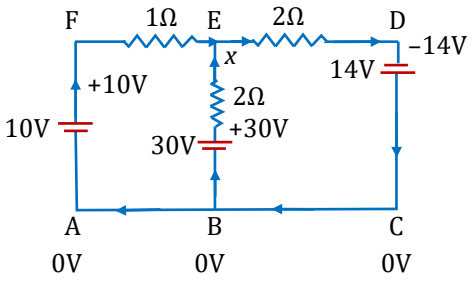
Let potential at $E = x$

Now applying Kirchhoff's current law at junction E.

(This can be applied at any other junction also).

$$\frac{x-10}{1}+\frac{x-30}{2}+\frac{x+14}{2}=0$$

$$4x = 36 \quad \Rightarrow x = 9$$



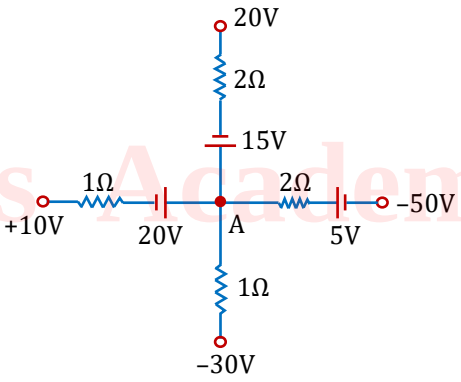
Current in $EF = \frac{10-9}{1} = 1\ A$ from F to E

Current in $BE = \frac{30-9}{2} = 10.5$ A from B to E

Current in $DE = \frac{9-(-14)}{2} = 11.5$ A from E to D

Illustration 25:

Find the potential at point A.



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Solution:

Let potential at $A = x$, applying Kirchhoff current law at junction A

$$\frac{x-20-10}{1}+\frac{x-15-20}{2}+\frac{x+45}{2}+\frac{x+30}{1}=0$$

$$\Rightarrow \quad \frac{2x-60+x-35+x+45+2x+60}{2}=0$$

$$\Rightarrow \quad 6x + 10 = 0$$

$$\Rightarrow \quad x = -5/3$$

Potential at $A = \frac{-5}{3}\ V$

Second law or loop rule

In any closed circuit the algebraic sum of e.m.f.'s and algebraic sum of potential drops is zero.

$$\sum IR + \sum E = 0$$

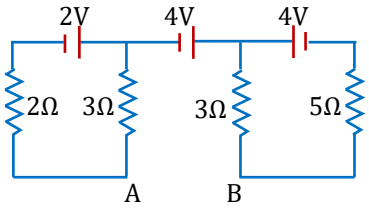
while moving from negative to positive terminal inside the cell. e.m.f. is taken as positive while moving in the direction of current in a circuit the potential drop (i.e. IR) across resistance is taken as negative.

This law is based on law of conservation of energy.

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Illustration 26:

In the given circuit calculate potential difference between A and B.



Solution:

First applying KVL on left mesh

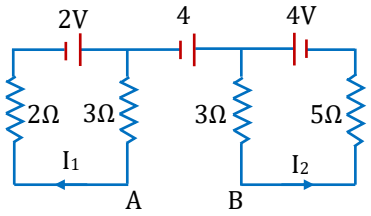
$2 - 3I_1 - 2I_2 = 0$

$I_1 = 0.4\ amp$

Now applying KVL on right mesh.

$4 - 5I_2 - 3I_2 = 0$

$I_2 = 0.5\ amp.$



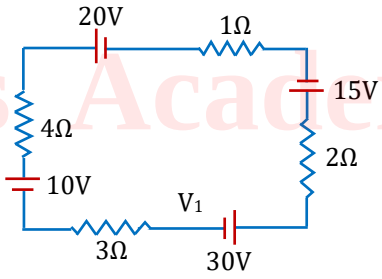
Potential difference between points A and B

$V_A + 3I_1 + 4 - 3I_2 = V_B$

$V_A - V_B = -3 \times 0.4 - 4 + 3 \times 0.5 = -3.7\ Volt.$

Illustration 27:

Find current in the circuit.



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Solution:

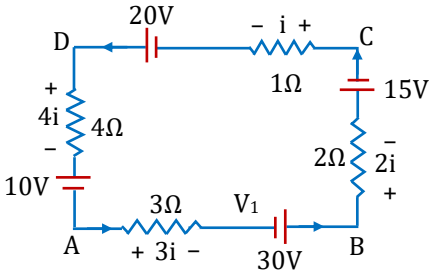
All the elements are connected in series current is all of them will be same let current = i

Applying Kirchhoff voltage law in $ADCBA$ loop

$10 + 4i - 20 + i + 15 + 2i - 30 + 3i = 0$

$10i = 25$

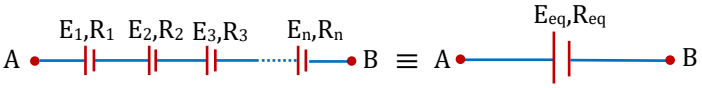
$i = 2.5\ A$



Combination of Cells

Grouping of Cells

Cells in Series:



Equivalent EMF

$E_{eq} = E_1 + E_2 + \ldots\ldots + E_n$ [write EMF's with polarity]

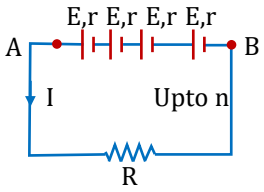
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Equivalent internal resistance

$$r_{eq} = r_1 + r_2 + r_3 + r_4 + \ldots + r_n$$

If n cells each of emf E and internal resistance r are connected in series with an external resistance R, then total emf = nE, so current in the circuit is

$$I = \frac{nE}{R + nr}$$



If $nr << R$ then $I = \frac{nE}{R}$ ⟶ Series combination is advantageous.

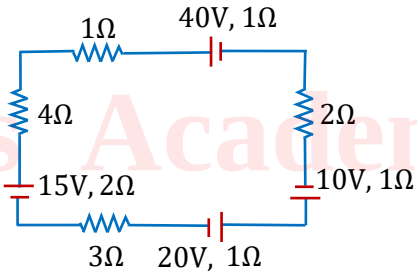
If $nr >> R$ then $I = \frac{E}{r}$ ⟶ Series combination is not advantageous.

Note:

If polarity of m cells is reversed, then equivalent emf = $(n\text{-}2m)\ E$ while the equivalent resistance is still $nr + R$, so current in R will be $i = \frac{(n-2m)E}{nr+R}$.

Illustration 28:

Find the current in the loop.



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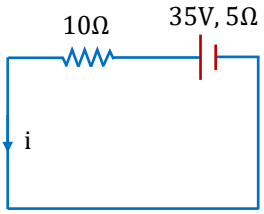
Solution:

The given circuit can be simplified as

$$\varepsilon_{eq} = 40 - 15 + 20 - 10 = 35V$$

$$r_{eq} = 1 + 2 + 1 + 1 = 5\Omega$$

$$R_{eq} = 1 + 4 + 3 + 2 = 10\Omega$$



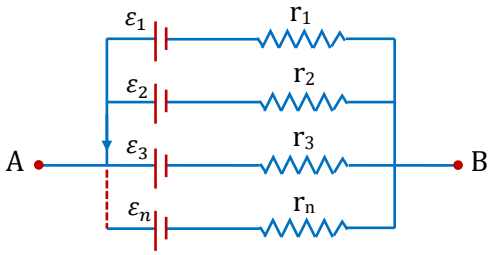
So $i = \frac{35}{10+5} = \frac{35}{15}$

$\Rightarrow$ $i = \frac{7}{3} A$

Cells in Parallel:

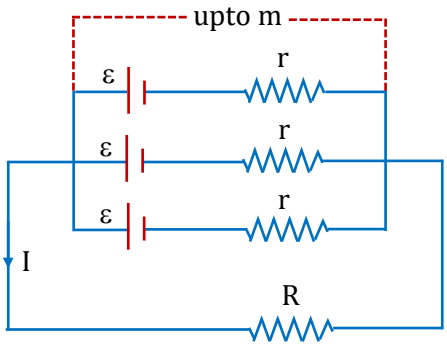
$$\varepsilon_{eq} = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2 + \ldots + \varepsilon_n/r_n}{1/r_1 + 1/r_2 + \ldots + 1/r_n}$$ [Use emf's with polarity]

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \ldots + \frac{1}{r_n}$$



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If m cells each of emf ε and internal resistance r be connected in parallel and if this combination is connected to an external resistance (R).



Then equivalent emf of the circuit = ε

Internal resistance of the circuit = $\frac{r}{m}$

and $I = \frac{\varepsilon}{R + \frac{r}{m}} = \frac{m\varepsilon}{mR + r}$

If $mR << r$

$I = \frac{m\varepsilon}{r}$ $\rightarrow$ Parallel combination is advantageous.

If $mR >> r$

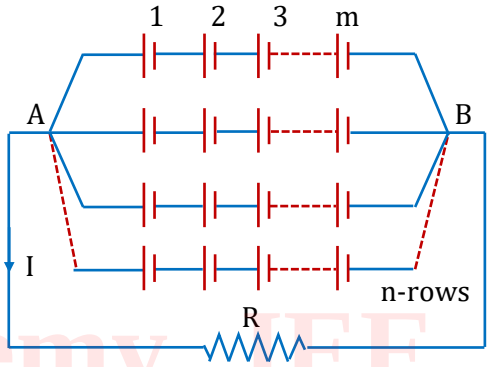
$I = \frac{\varepsilon}{R}$ $\rightarrow$ Parallel combination is not advantageous.

Mixed combination:

mn = number of identical cells

n = number of rows

m = number of cells in each row



If internal resistance of each cell is r, then the combination of cells is equivalent to single cell of

$emf = mE$

and Internal resistance = $\frac{mr}{n}$

Current, $I = \frac{mE}{R + \frac{mr}{n}}$

For maximum current $nR = mr$

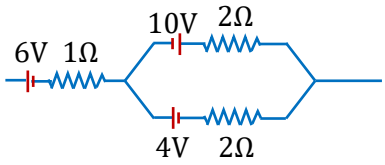
or $R = \frac{mr}{n}$ = internal resistance of the equivalent battery.

$I_{max} = \frac{nE}{2r} = \frac{mE}{2R}$

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Illustration 29:

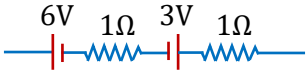
Find the emf and internal resistance of a single battery which is equivalent to a combination of three batteries as shown in figure.



Solution:

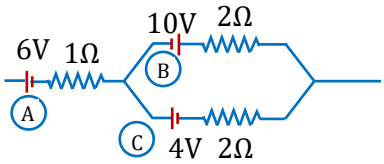
Battery (B) and (C) are in parallel combination with opposite polarity. So, their equivalent

$\varepsilon_{BC} = \frac{\frac{10}{2} + \frac{-4}{2}}{\frac{1}{2} + \frac{1}{2}} = \frac{5-2}{1} = 3V, r_{BC} = 1\Omega$



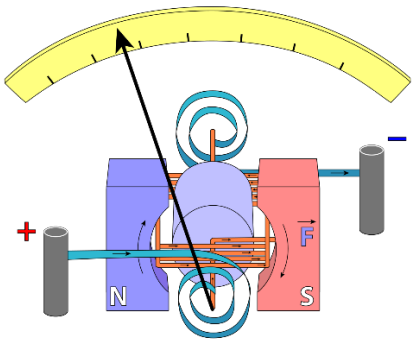
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Now,



$\varepsilon_{ABC} = 6-3 = 3V,\ r_{ABC} = 2\Omega$

Galvanometer



The instrument used to measure strength of current, by measuring the deflection of the coil due to torque produced by a magnetic field, is known as galvanometer.

It consists of a pivoted coil placed in the magnetic field of a permanent magnet. A spring is attached to the coil. In the equilibrium position, with no current in the coil, the pointer is at zero and spring is relaxed. When there is a current in the coil, the magnetic field exerts a torque on the coil that is proportional to current. As the coil turns, the spring exerts a restoring torque that is proportional to the angular displacement. Thus, the angular deflection of the coil and pointer is directly proportional to the coil current and the device can be calibrated to measure current.



$\boxed{I \propto \phi}$

Galvanometer is represented in circuit as follows:



The rate of variation of deflection depends upon the magnitude of deflection itself and so the accuracy of the instrument.

- **Current sensitivity**

  The ratio of deflection to the current i.e. deflection per unit current is called current sensitivity (CS) of the galvanometer

  $$CS = \frac{\phi}{I}$$

- **Voltage sensitivity**

  The ratio of deflection to potential difference i.e deflection per unit potential difference is called voltage sensitivity (VS) of the galvanometer

  $$VS = \frac{\phi}{V}$$

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Shunt

The small resistance connected in parallel to galvanometer coil, in order to control current flowing through the galvanometer, is known as shunt.

Merits of shunt:

(i) To protect the galvanometer coil from burning.

(ii) Any galvanometer can be converted into ammeter of desired range with the help of shunt.

(iii) The range of an ammeter can be changed by using shunt resistance of different values.

Demerits of shunt:

Shunt resistance decreases the sensitivity of galvanometer.

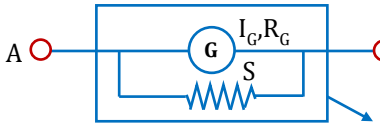
Conversion of galvanometer into ammeter

A galvanometer can be converted into an ammeter by connecting low resistance in parallel to its coil.

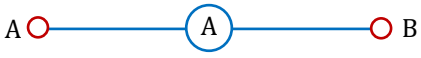


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Ammeter is represented as follow -



Whole system is known as Ammeter



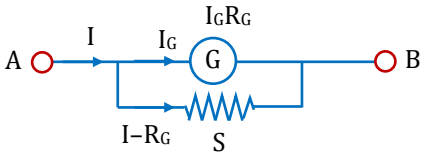
If maximum value of current to be measured by ammeter is I then

$$I_G \cdot R_G = (I - I_G)S$$

$$S = \frac{I_G.R_G}{I - I_G}$$

where I = Maximum current that can be measured using the given ammeter.

For measuring the current the ammeter is connected in series.



In calculation it is simply a resistance

Resistance of ammeter $R_A = \frac{R_G.S}{R_G + S}$

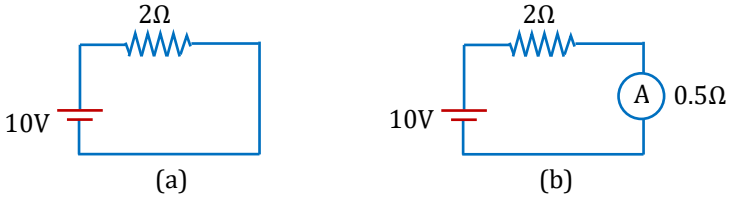
Note:

An ideal ammeter has zero resistance.

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Illustration 30:

Find the current in the circuit (a) and (b) and also determine percentage error in measuring the current through an ammeter.



Solution:

Current in circuit (a) $I = \frac{10}{2} = 5A$

Current in circuit (b) $I' = \frac{10}{2.5} = 4A$

Percentage error in measurement of current $= \frac{I - I'}{I} \times 100 = 20\%$

Here we see that due to ammeter the current has reduced. A good ammeter has very low resistance as compared with other resistors, so that due to its presence in the circuit the current is not affected.

Illustration 31:

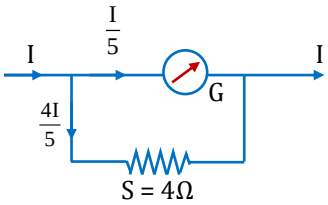
When a shunt of 4Ω is attached to a galvanometer, the deflection reduces to $1/5^{th}$. If an additional shunt of 2Ω is attached what will be the deflection?

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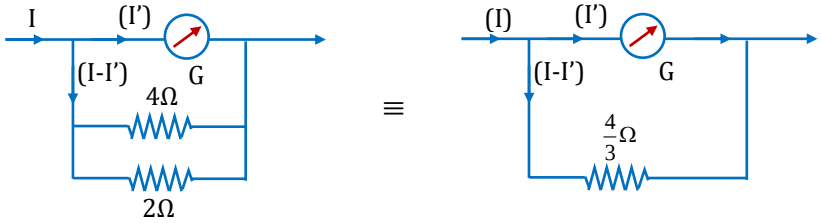
Solution:

Initial condition: When shunt of 4Ω is used,

$$\frac{I}{5} \times G = \frac{4}{5} I \times 4 \Rightarrow G = 16\Omega$$



When additional shunt of 2Ω is used,



$$I' \times 16 = \frac{4}{3}(I - I') \Rightarrow I' = \frac{I}{13}$$

$\therefore$ It will reduce to $\frac{I}{13}$ of the initial deflection.

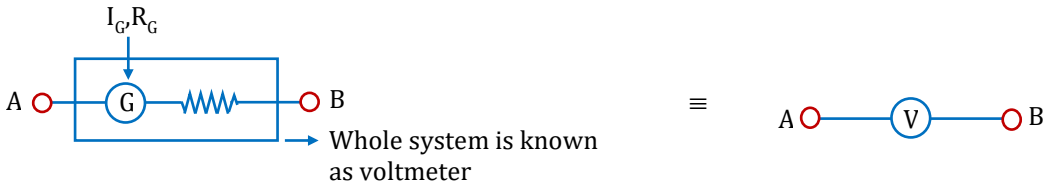
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Conversion of galvanometer into voltmeter:

The galvanometer can be converted into voltmeter by connecting high resistance in series with its coil.



It is used to measure potential difference across a resistor in a circuit.



For maximum potential difference,

$$V = I_G R + I_G R_G$$

$$R = \frac{V}{I_G} - R_G$$

For calculation it is simply a resistance



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Resistance of voltmeter,

$$R_V = R_G + R$$

$$I_G = \frac{V}{R_G + R}$$

Note:

A good voltmeter has high value of resistance.

$$R_V = R_G + R \simeq R$$

Ideal voltmeter $\Rightarrow R \to \infty$

Illustration 32:

A galvanometer having 30 divisions has current sensitivity of 20 μA/division. It has a resistance of 25Ω.

(i) How will you convert this galvanometer into an ammeter of range 1 ampere?

(ii) How will you convert this galvanometer into a voltmeter of range 1 volt?

Solution:

The current required for full scale deflection, $I_g = 20\mu A \times 30 = 600\mu A = 6 \times 10^{-4} A$

(i) To convert it into ammeter, a shunt is required in parallel with it

Shunt resistance $R_s' = \frac{I_g R_g}{(I - I_g)} = \left(\frac{6\times10^{-4}}{1-6\times10^{-4}}\right) 25 = 0.015\Omega$

(ii) To convert galvanometer in voltmeter, a high resistance is required in series with it

Series resistance $R = \frac{V}{I_g} - R_g = \frac{1}{6\times10^{-4}} - 25 = 1666.67 - 25 = 1641.67\Omega$

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Verification of Ohm's Law Experiment

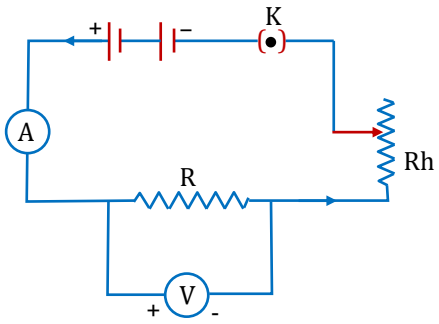
Verification of Ohm's law using voltmeter and ammeter

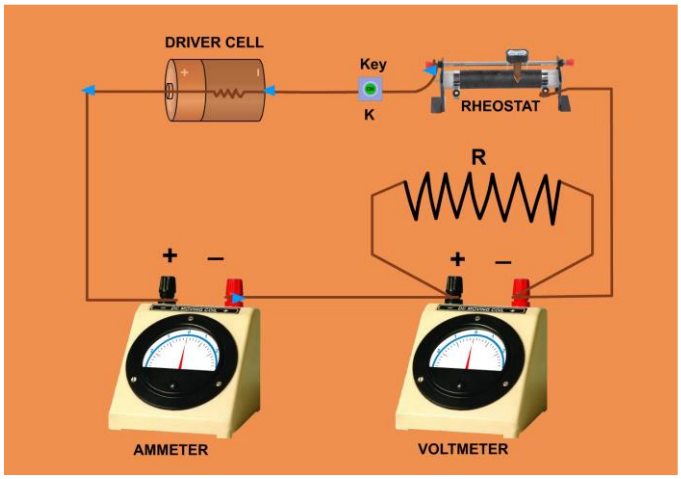
Ohm's law states that the electric current I flowing through a conductor is directly proportional to the potential difference (V) across its ends provided that the physical conditions of the conductor (such as temperature, dimensions, etc.) are kept constant. Mathematically,

$V \propto I$ or $V = IR$

Here R is a constant known as resistance of the conductor and depends on the nature and dimensions of the conductor.

Circuit Diagram: The circuit diagram is as shown below:





Procedure: By shifting the rheostat contact, reading of ammeter and voltmeter are noted down. At least six set of observations are taken. Then a graph is plotted between potential difference (V) across R and current (I) through R. The graph comes to be a straight line as shown in figure.

Result: It is found from the graph that the ratio V/I is constant. Hence, current voltage relationship is established, i.e. $V \propto I$. It means Ohm's law is established.

As $I = \left(\frac{1}{R}\right) V$, find the slope of $I - V$ curve and equate it to $\frac{1}{R}$.

$$\text{Slope} = \frac{BP}{AP} = \frac{1}{R}$$

Get $$R = \frac{1}{\text{Slope of } I-V \text{ graph}}$$

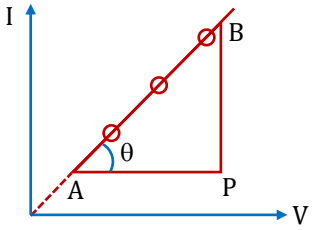
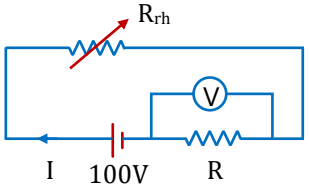
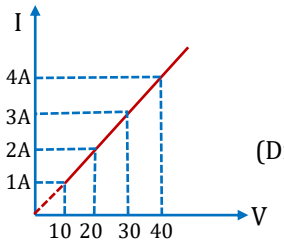


Illustration 33:

If emf of battery is 100 V, then what was the resistance of Rheostat adjusted at 2nd reading ($I = 2A$, $V = 20V$).





(Drawn across R)

(A) 10Ω (B) 20Ω (C) 30Ω (D) 40Ω

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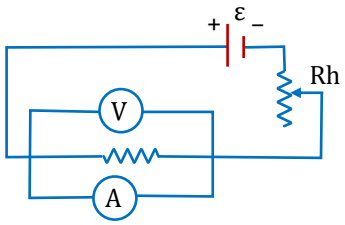
Solution:

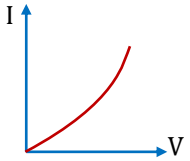
From the curve slope $= \frac{I}{V} = \frac{1}{R} = \frac{1}{10}$; $R = 10\ \Omega$

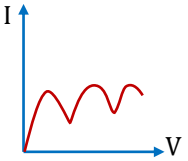
For second reading, $I = \frac{Emf}{R+R_{rh}} \Rightarrow 2 = \frac{100}{10+R_{rh}} \Rightarrow R_{rh} = 40\Omega$

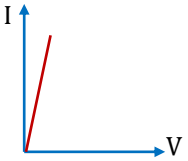
Illustration 34:

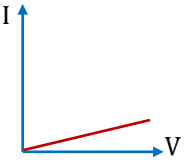
If by mistake, Ammeter is connected parallel to the resistance then *I-V* curve expected is (Here *I* = reading of ammeter, *V* = reading of voltmeter)



(A) 

(B) 

(C) 

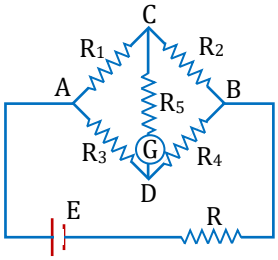
(D) 

Solution:

As ammeter has very low resistance most of current will pass through the ammeter so reading of ammeter (*I*) will be very large. Voltmeter has very high resistance so reading of voltmeter will be very low. Therefore, most appropriate answer is option C.

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Wheatstone Bridge



The arrangement as shown in figure of five resistors is known as Wheatstone bridge.

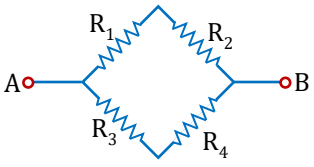
Here there are four terminals in which except two all are connected to each other through resistive elements.

In this circuit

If $R_1R_4 = R_2R_3$ or $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ then $V_C = V_D$ and current in $R_5 = 0$. This is called balance point or null point i.e. current through the galvanometer is zero.

Here in this case products of opposite arms are equal.

Potential difference between *C* and *D* at null point is zero and wheatstone arrangement is said to be balanced. Hence, here the circuit can be assumed to be following for calculating R_{eq} across *AB*.

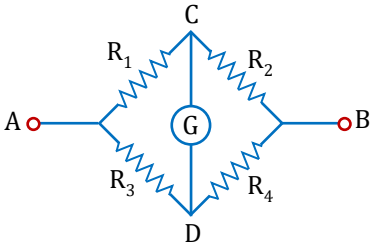


Note:

The null point is not affected by resistance R_5, *E* and *R*.

It is not affected even if the positions of Galvanometer and battery (*E*) are interchanged.

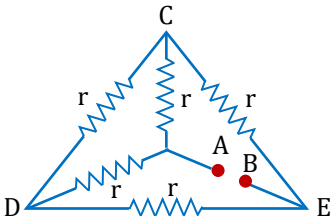
- If $\frac{R_1}{R_2} < \frac{R_3}{R_4}$ then $V_C > V_D$ and current will flow from *C* to *D*.
- If $\frac{R_1}{R_2} > \frac{R_3}{R_4}$ then $V_C < V_D$ and current will flow from *D* to *C*.



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Illustration 35:

In the adjoining network of resistors each is of resistance r Ω. Find the equivalent resistance between point A and B.



Solution:

Given circuit is balanced Wheatstone Bridge.

$$\therefore \quad \frac{1}{R_{AB}} = \frac{1}{2r} + \frac{1}{2r} = \frac{1}{r}$$

$$\therefore \quad R_{AB} = r$$

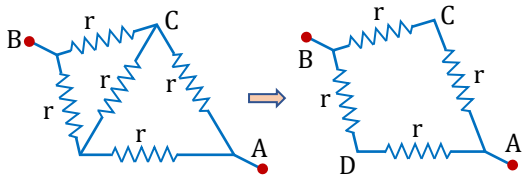


Illustration 36:

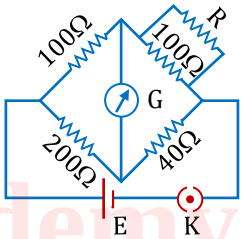
For the following diagram the galvanometer shows zero deflection then what is the value of R?

Solution:

For balanced Wheatstone bridge

$$\therefore \quad \frac{100}{\frac{100R}{(100+R)}} = \frac{200}{40} \quad \Rightarrow \quad \frac{100+R}{R} = 5$$

$$100 + R = 5R \quad \Rightarrow \quad R = \frac{100}{4} = 25\Omega$$



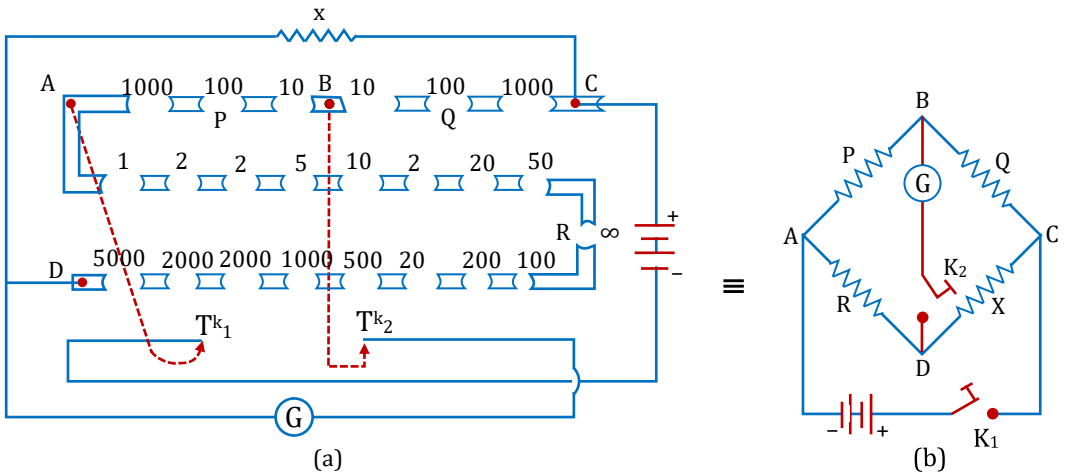
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Post Office Box

A post office box is a compact form of Wheatstone bridge with the help of which we can measure the value of the unknown resistance correctly up to 2nd decimal place, i.e., up to 1/100th of an ohm correctly.

In figure (a), each of the arms AB and BC contains three resistances of 10, 100 and 1000 ohm. These arms are called the ratio arms. The third arm AD, called the resistance arm, is a complete resistance box containing resistances from 1Ω to 5,000 Ω. The unknown resistance X constitutes the fourth arm CD. Thus, the four arms AB, BC, CD and AD are infact the four arms of the Wheatstone bridge (figure (b)).

A cell is connected between A and C with key K_1 and Galvanometer is connected between B and D with key K_2.



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Working: The working of the post office box involves broadly the following four steps:

I. Keeping R zero, each of the resistances P and Q are made equal to 10 ohm by taking out suitable plugs from the arms AB and BC respectively. After pressing the battery key first and then the galvanometer key, the direction of deflection of the galvanometer coil is noted. Now, making R infinity, the direction of deflection is again noted. If the direction is opposite to that in the first case, then the connections are correct.

II. Keeping both P and Q equal to 10Ω, the value of R is adjusted, beginning from 1Ω, till 1Ω increase reverses the direction of deflection. The 'unknown' resistance clearly lies somewhere between the two final values of R.

$$\left[X = R\frac{Q}{P} = R\frac{10}{10} = R\right]$$

As an illustration, suppose with 4Ω resistance in the arm AD, the deflection is towards left and with 5Ω, it is towards right. The unknown resistance lies between 4Ω and 5Ω.

III. Making P 100 Ω and keeping Q 10 Ω, we again find those values of R between which direction of deflection is reversed. Clearly, the resistance in the arm AD will be 10 times the resistance X of the wire.

$$\left[X = R\frac{Q}{P} = R\frac{10}{100} = \frac{R}{10}\right]$$

In the illustration considered in step II, the resistance in the arm AD will now lie between 40 Ω, and 50 Ω. So, in this step, we have to start adjusting R from 40 Ω onwards. If 42 Ω and 43 Ω are the two values of R which give opposite deflections, then the unknown resistance lies between 4.2 Ω and 4.3 Ω.

IV. Now, P is made 1000 Ω and Q is kept at 10 Ω. The resistance in the arm AD will now be 100 times the 'unknown' resistance.

$$\left[X = R\frac{10}{1000} = \frac{R}{100}\right]$$

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In the illustration under consideration, the resistance in the arm AD will lie between 420 Ω and 430 Ω. Suppose the deflection is to the right for 426 ohm, towards left for 424 ohm and zero deflection for 425 Ω then, the unknown resistance is 4.25 Ω.

No. of Obs.	Resistance in the Ratio arms		Resistance in arm AD (R) (Ohm)	Direction of deflection left or right	Unknown resistance $X = \frac{Q}{P} \times R$ (Ohm)
	AB (P) (Ohm)	BC(Q) (Ohm)			
1.	10	10	4	left	(4–5)
			5	Right	
2.	100	10	40	Left (large)	(4.2 – 4.3)
			50	Right (large)	
			42	Left	
			43	Right	
3.	1000	10	420	Left	4.25
			424	Left	
			425	No deflection	
			426	Right	

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Illustration 37:

While using a post office box the keys should be switched on in the following order:

(A) First cell key and then galvanometer key. (B) First the galvanometer key and then cell key.

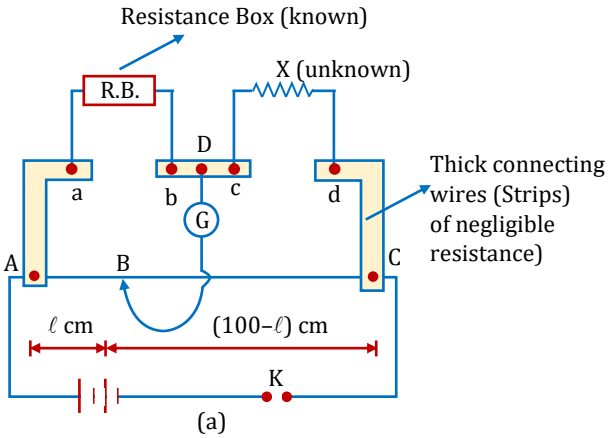
(C) Both the keys simultaneously. (D) Any key first and then the other key.

Solution:

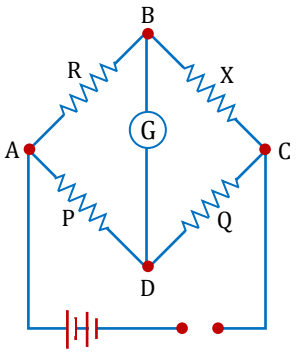
First cell key and then galvanometer key.

Meter Bridge

- It is based on principle of Wheatstone bridge.
- It is used to find the value of unknown resistance of wire.



(a)



(b)

If $AB = \ell$ cm, then $BC = (100 - \ell)$ cm.

Resistance P of the wire between A and B will be proportional to ℓ i.e., $P \propto \ell$

[∵ Specific resistance ρ and cross-sectional area A are same for whole of the wire]

Similarly, if Q is resistance of the wire between B and C, then

$$Q \propto 100-\ell$$

So, $$\frac{P}{Q} = \frac{\ell}{100-\ell}$$

Applying the condition for balanced Wheatstone bridge, we get

$$RQ = PX$$

$$\therefore \quad X = R\frac{Q}{P} \quad \text{or} \quad X = \frac{100-\ell}{\ell}R$$

Since R and ℓ are known, therefore, the value of X can be calculated.

Note:

For better accuracy, R is so adjusted that ℓ lies close to 50 cm.

End Corrections

In meter Bridge circuit, some extra length comes (is found under metallic strips) at end point A and C. So, some additional length (α and β) should be included at ends for accurate result. Hence in place of ℓ we use $\ell + \alpha$ and in place of $100 - \ell$, we use $100 - \ell + \beta$ (where α and β are called end correction). To estimate α and β, we use known resistance R_1 and R_2 at the place of R and X in meter Bridge.

Suppose we get null point at ℓ_1 distance then

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$$\frac{R_1}{R_2}=\frac{\ell_1+\alpha}{100-\ell_1+\beta} \quad \text{...(i)}$$

Now we interchange the position of R_1 and R_2, and get null point at ℓ_2 distance then

$$\frac{R_2}{R_1}=\frac{\ell_2+\alpha}{100-\ell_2+\beta} \quad \text{...(ii)}$$

Solving equation (i) and (ii) get,

$$\alpha=\frac{R_2\ell_1-R_1\ell_2}{R_1-R_2} \text{ and } \beta=\frac{R_1\ell_1-R_2\ell_2}{R_1-R_2}-100$$

These end corrections (α and β) are used to modify the observations.

Illustration 38:

If we used 100Ω and 200Ω resistance in place of R and X, we get null deflection at ℓ_1 = 33.0cm. If we interchange the Resistance, the null deflection was found to be at ℓ_2 = 67.0 cm. The end correction α and β should be :

(A) α = 1cm, β = 1cm
(B) α = 2cm, β = 1cm
(C) α = 1cm, β = 2cm
(D) None of these

Solution:

$$\alpha=\frac{R_2\ell_1-R_1\ell_2}{R_1-R_2}=\frac{(200)(33)-(100)(67)}{100-200}=1 \text{ cm}$$

$$\beta=\frac{R_1\ell_1-R_2\ell_2}{R_1-R_2}-100=\frac{(33)(100)-(200)(67)}{100-200}-100=1cm$$

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Potentiometer

Necessity of potentiometer

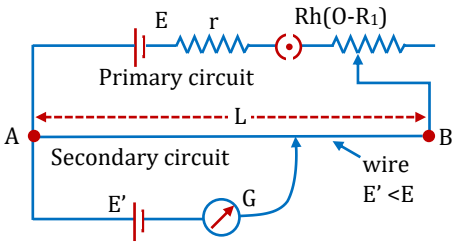
Practically voltmeter has finite resistance. (ideally it should be ∞) in other words it draws some current from the circuit. To overcome this problem potentiometer is used because at the instant of measurement, it does not draws current from the circuit.

A potentiometer is a linear conductor of uniform cross-section with a steady current set up in it. This maintains a uniform potential gradient along the length of the wire. Any potential difference which is less than the potential difference maintained across the potentiometer wire can be measured using this.

The wire should have high resistivity and low expansion coefficient.

For example: Manganin or, Constantine wire etc.

Circuit of Potentiometer



- **Primary circuit** contains constant source of voltage rheostat or Resistance Box.
- **Secondary circuit** contains galvanometer and unknown cell.

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- **Potential gradient (x) (V/m)**

  The fall of potential per unit length of potentiometer wire is called potential gradient.

$$x = \frac{V}{L} = \frac{\text{current} \times \text{resitance of potentiometer wire}}{\text{length of potentiometer wire}} = I\left(\frac{R_p}{L}\right)$$

Illustration 39:

A Potentiometer wire of 10 m length and having 10 ohm resistance, emf 2 volts and a rheostat. If the potential gradient is 1 micro volt/mm, the value of resistance in rheostat in ohms will be:

(A) 1.99 (B) 19.9 (C) 199 (D) 1990

Solution:

ℓ = 10 m, R = 10Ω,

$E = 2$ volts, $\frac{dV}{d\ell} = 1\mu V/mm$

$$\frac{dV}{d\ell} = \frac{1\times10^{-6}}{1\times10^{-3}} V/m = 1 \times 10^{-3} V/m$$

Across wire potential drop, $\frac{dV}{d\ell} \times \ell = 1 \times 10^{-3} \times 10 = 0.01$ volts

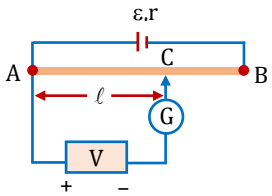
$$I = \frac{0.01}{10} = 0.001 = \frac{E}{R+R'} (R' = \text{resistance of rheostat})$$

$$R' = \frac{E}{0.001} - R = \frac{2}{0.001} - 10 = 2000 - 10 = 1990\Omega$$

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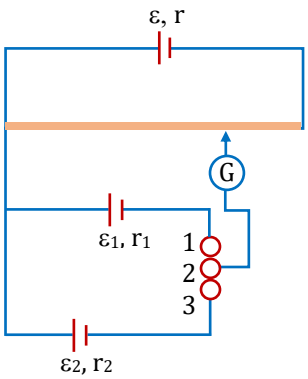
Application of Potentiometer

(A) To measure unknown emf of cell



The unknown voltage V is connected across the potentiometer wire as shown in figure. The positive terminal of the unknown voltage is kept on the same side as of the source of the top most battery. When reading of galvanometer is zero then we say that the meter is balanced. In that condition $V = x\ell$.

(B) To find emf of unknown cell and compare emf of two cells



In case I, In figure, (2) is joint to (1) then balance length = ℓ_1

$$\varepsilon_1 = x\ell_1 \quad \text{...(1)}$$

In case II, In figure, (3) is joint to (2)

Then balance length = ℓ_2

$$\varepsilon_2 = x\ell_2 \quad \text{...(2)}$$

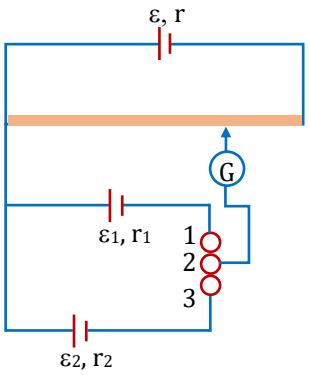
$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{\ell_1}{\ell_2}$$

If any one of ε_1 or ε_2 is known the other can be found. If x is known then both ε_1 and ε_2 can be found.

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Illustration 40:

In an experiment to determine the emf of an unknown cell, its emf is compared with a standard cell of known emf $\varepsilon_1 = 1.12\ V$. The balance point is obtained at 56 cm with standard cell and 80 cm with the unknown cell. Determine the emf of the unknown cell.



Solution:

Here, $\varepsilon_1 = 1.12\ V$; $\ell_1 = 56$ cm ; $\ell_2 = 80$ cm

Using equation

$\varepsilon_1 = x\ell_1$ ...(1)

$\varepsilon_2 = x\ell_2$ ...(2)

We get, $\frac{\varepsilon_1}{\varepsilon_2} = \frac{\ell_1}{\ell_2} \Rightarrow \varepsilon_2 = \varepsilon_1\left(\frac{\ell_2}{\ell_1}\right)$ or $\varepsilon_2 = 1.12\left(\frac{80}{56}\right) = 1.6V$ **Ans.**

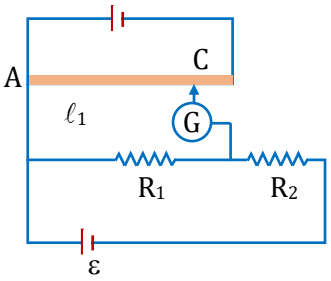
(C) To find current if resistance is known

$V_A - V_C = x\ell_1$

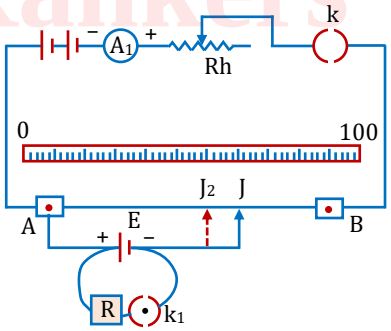
$IR_1 = x\ell_1$; $I = \frac{x\ell_1}{R_1}$

Similarly, we can find the value of R_2 also.

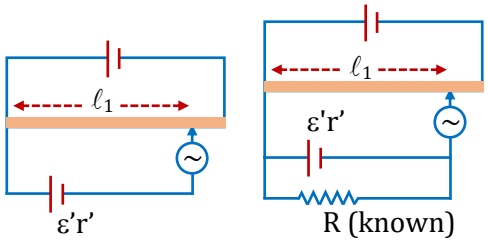
Potentiometer is ideal voltmeter because it does not draw any current from circuit, at the balance point.



(D) To find the internal resistance of cell



1st arrangement



2nd arrangement

By first arrangement, $\varepsilon' = x\ell_1$ ...(1)

By second arrangement, $IR = x\ell_2$

$I = \frac{x\ell_2}{R}$

Also $I = \frac{\varepsilon'}{r'+R}$

$\Rightarrow \frac{\varepsilon'}{r'+R} = \frac{x\ell_2}{R}$

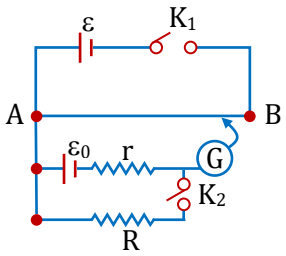
$\Rightarrow \frac{x\ell_1}{r'+R} = \frac{x\ell_2}{R}$

$\Rightarrow r' = \left[\frac{\ell_1 - \ell_2}{\ell_2}\right]R$

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Illustration 41:

The internal resistance of a cell is determined by using a potentiometer. ln an experiment, an external resistance of 60Ω is used across the given cell. When the key is closed, the balance length on the potentiometer decreases from 72 cm to 60 cm. calculate the internal resistance of the cell.



Solution:

According to equation,

$\varepsilon_0 = x\ell_0$ ...(i)

$V = IR = x\ell$ ...(ii)

$I = \frac{\varepsilon_0}{R+r}$ ...(iii)

From equation (i), (ii) and (iii) we get

$$r = R\left(\frac{\ell_0}{\ell} - 1\right)$$

Here ℓ_0 = 72cm; ℓ = 60 cm; R = 60 Ω

$\therefore \quad r = (60)\left(\frac{72}{60} - 1\right)$ or $r = 12\Omega$.

Note:

(a) If same polarity of 1 cell and 2 cell are not connected with each other then balancing condition (null deflection) cannot be obtained.

(b) If potential drop across wire is less than potential drop of secondary circuit then balancing condition cannot be obtained.

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Difference between potentiometer and voltmeter	
Potentiometer	**Voltmeter**
• It measures the unknown emf very accurately • While measuring potential difference the resistance of potentiometer becomes infinite. • It is based on zero deflection method. • It has a high sensitivity. • It is used for various applications like measurement of internal resistance of cell, calibration of ammeter and voltmeter, measurement of thermo emf, comparison of emfs etc.	• It measures the unknown emf approximately. • While measuring emf it draws some current from the source of emf. • Its sensitivity is low. • While measuring unknown potential difference the resistance of voltmeter is high but finite. • It is based on deflection method. • It is only used to measure unknown potential difference.

Sensitivity:

It is the ability of potentiometer to read very small value of potential difference easily.

So, sensitivity is inversely proportional to potential gradient of potentiometer wire.

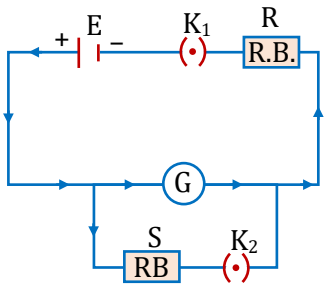
$$\text{Sensitivity} \propto \frac{1}{P.G.} \propto \frac{\ell}{V} \propto \frac{\text{Length of wire}}{\text{potential drop across wire}}$$

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Half Deflection Method

Aim:

To find the resistance of a galvanometer by half deflection method and find its figure of merit.



Theory:

The connections for finding the resistance of a galvanometer by the half deflection method are shown in Fig. When the key K_1 is closed, keeping the key K_2 open, the current I_g through the galvanometer is given by

$I_g = \frac{E}{R+G}$, where E = *E.M.F.* of the cell.

R = Resistance from the resistance box R.B.

G = Galvanometer resistance.

If θ is the deflection produced, then

$$\frac{E}{R+G} = k\theta \qquad ...(1)$$

If now the key K_2 is closed and the value of the shunt resistance S is adjusted so that the deflection is reduced to half of the first value, then current flowing through the galvanometer I'_g is given by

$$I'_g = \frac{E}{R + \frac{GS}{(G+S)}}\left(\frac{S}{G+S}\right) = \frac{k\theta}{2}$$

or $$I'_g = \frac{ES}{R(G+S)+GS} = \frac{k\theta}{2} \qquad ...(2)$$

Comparing (1) and (2), we get

$$(R+G)2S = R(G+S) + GS$$

or $$(R-S)G = RS \quad \text{or} \quad G = \frac{RS}{R-S}$$

If the value of R is very large as compared to S, then $\frac{R}{R-S}$ is nearly equal to unity. Hence

$$\boxed{G \approx S}$$

Figure of Merit:

Figure of merit of a galvanometer is that much current sent through the galvanometer in order to produce a deflection of one division on the scale.

If k is the figure of merit of the galvanometer and 'θ' be the number of divisions on the scale, then current (I_g) through the galvanometer is given by

$$\boxed{I_g = k\theta}$$

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Precautions:

1. The value of '*R*' should be large
2. To decrease the deflection, the shunt resistance should be decreased and vice-versa.
3. In this method it is assumed that the deflection is proportional to the current. This is possible only in a Weston type moving coil galvanometer.
4. The connections must be tight and the ends of connecting wires should be cleaned.

Illustration 42:

The series combination of variable resistance *R*, one 100 Ω resistor and moving coil galvanometer is connected to a mobile phone charger having negligible internal resistance. The zero of the galvanometer lies at the centre and the pointer can move 30 division full scale on either side depending on the direction of current. The reading of the galvanometer is 10 divisions and the voltages across the galvanometer and 100 Ω resistor are respectively 12 mV and 16 mV.

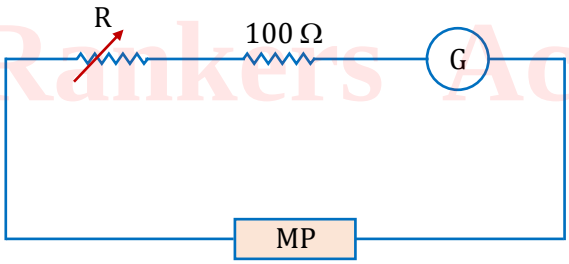
1. The figure of merit of the galvanometer in microampere per division is:

(A) 16 (B) 20 (C) 32 (D) 10

2. The resistance of the galvanometer (in Ω) is:

(A) 50 Ω (B) 75 Ω (C) 100 Ω (D) 80 Ω

Solution:



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$10I_0G = 12 \times 10^{-3}$

$10I_0(100) = 16 \times 10^{-3} \Rightarrow 10^3 I_0 = 16 \times 10^{-3} \Rightarrow I_0 = 16 \times 10^{-6} A$

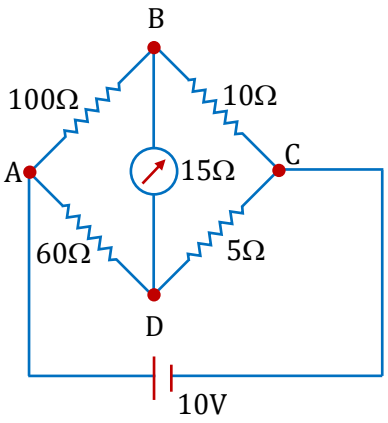
$\frac{G}{100} = \frac{3}{4} \Rightarrow G = 75\Omega$

Resistance Calculation of Complex Circuits

Unbalanced Wheatstone Bridge

Illustration 43:

The four arms of a Wheatstone bridge (Fig.) have the following resistances: *AB* = 100Ω, *BC* = 10Ω, *CD* = 5Ω, and *DA* = 60Ω. A galvanometer of 15Ω resistance is connected across *BD*. Calculate the current through the galvanometer when a potential difference of 10V is maintained across *AC*.



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Solution:

Consider the mesh BADB, we have

$$100I_1 + 15I_g - 60I_2 = 0$$

Or $$20I_1 + 3I_g - 12I_2 = 0 \quad ...(1)$$

Considering the mesh BCDB, we have

$$10(I_1 - I_g) - 15I_g - 5(I_2 + I_g) = 0$$

$$10I_1 - 30I_g - 5I_2 = 0$$

$$2I_1 - 6I_g - I_2 = 0 \quad ...(2)$$

Considering the mesh ADCEA,

$$60I_2 + 5(I_2 + I_g) = 10$$

$$13I_2 + I_g = 2 \quad ...(3)$$

Multiplying equation (2) by 10

$$20I_I - 60I_g - 10I_2 = 0 \quad ...(4)$$

From equations (4) and (1) we have

$$63I_g - 2I_2 = 0$$

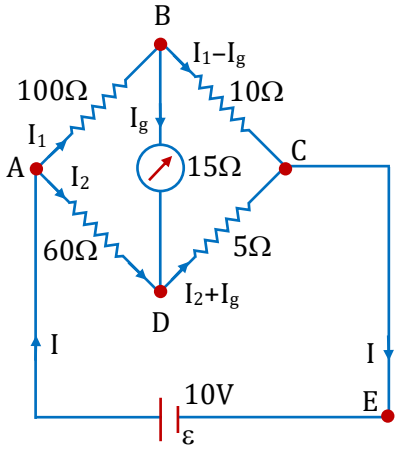
$$I_2 = 31.5I_g \quad ...(5)$$

Substituting the value of I_2 into equation (3), we get

$$13(31.5I_g) + I_g = 2$$

$$410.5I_g = 2$$

$$I_g = 4.87mA$$



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Symmetrical Circuits:

Some circuits can be modified to have simpler solution by using symmetry rules. One should note that if these are solved by traditional method of KVL and KCL then it would take much more time.

Mirror symmetry:

(1) It is applied along perpendicular bisector of line AB (A and B are the points about which $R_{eq.}$ is to be calculated)

(2) We can remove resistances and junctions which are on the line of symmetry.

(3) Point on line of symmetry are equipotential points.

Folding symmetry:

(1) It is applied along line AB.

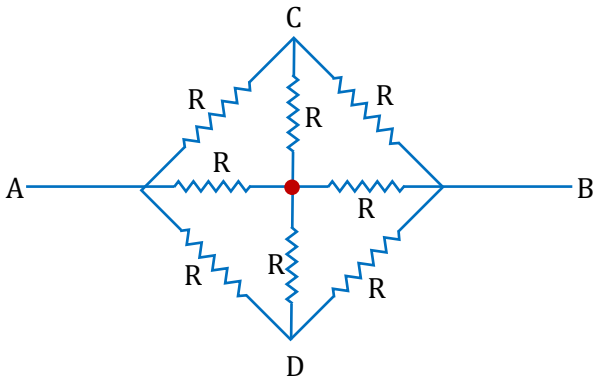
(2) We can fold the circuit about line of symmetry

(3) When we fold the circuit the parallel combination of resistance can be solved easily.

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Illustration 44:

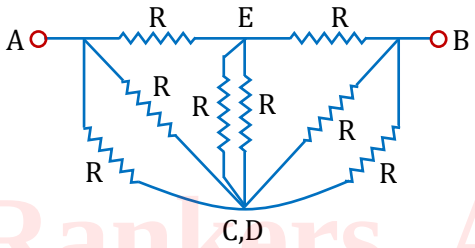
Find the equivalent Resistance between A and B



Solution:

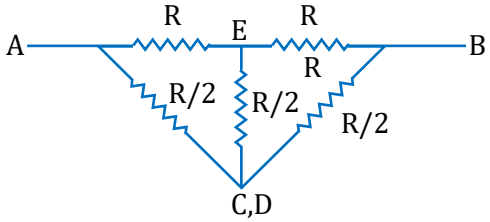
Let's use folding symmetry about line AB.

Circuit can be simplified as per folding symmetry.



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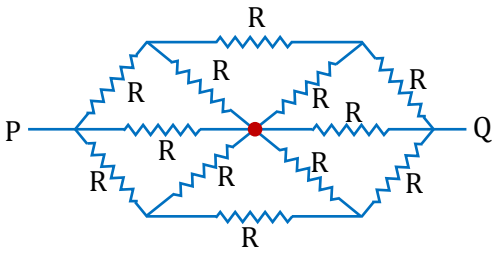
Now, it is Balanced Wheatstone bridge



$$R_{eq} = \frac{2R \times R}{2R + R} = \frac{2R}{3}$$

Illustration 45:

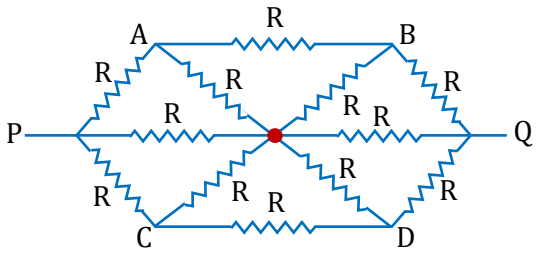
Find the equivalent Resistance between P and Q



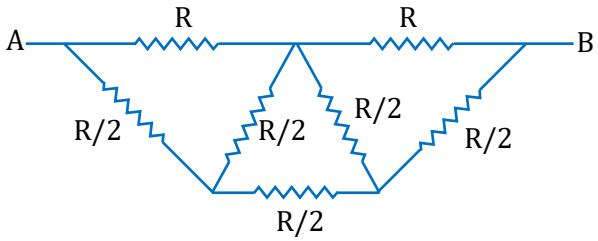
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Solution:

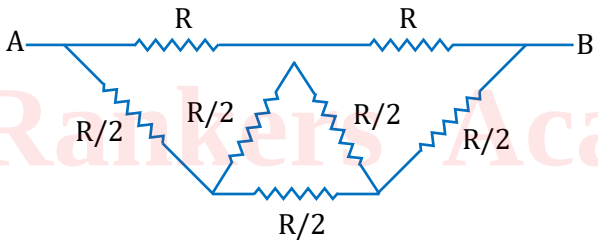
Here both folding and mirror symmetries exists. Let's start solving using folding symmetry.



Here $V_A = V_C$ and $V_B = V_D$

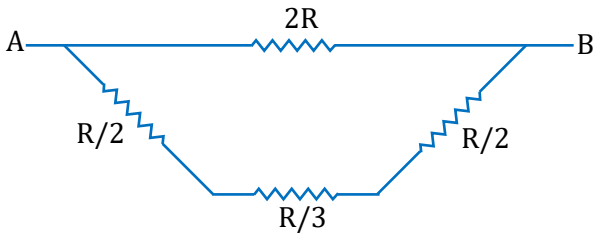


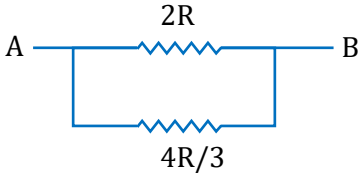
Here the circuit can be further simplified using mirror symmetry and junction can be removed.



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This circuit can be simplified as





$$R_{eq} = \frac{2R \times \frac{4R}{3}}{\frac{10R}{3}} = \frac{4R}{5}$$

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